Giới thiệu công nghệ giải trình tự thế hệ mới (NGS) và ứng dụng

04.09.2026

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Viện Nghiên cứu Ứng dụng Khoa học Sức khỏe và Lão hóa – Bệnh viện Thống Nhất

Nội dung

1. Giới thiệu dự án **BỘ GEN NGƯỜI & GIẢI TRÌNH TỰ GEN THẾ HỆ MỚI (NGS)**
2. Ví dụ mối quan hệ của **BIẾN THỂ GEN** và **BỆNH DI TRUYỀN**
3. Quy trình **XÉT NGHIỆM** gen bằng phương pháp giải trình tự thế hệ mới (NGS)
4. Năm ví dụ về **XÉT NGHIỆM** gen cho **BỆNH DI TRUYỀN** và **UNG THƯ**
5. Ứng dụng NGS trong bệnh nhiễm

Xin gởi lời cảm ơn trân thành đến

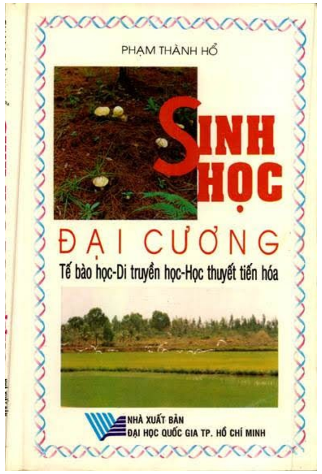
Bệnh viện Thống Nhất và Viện ARiHA

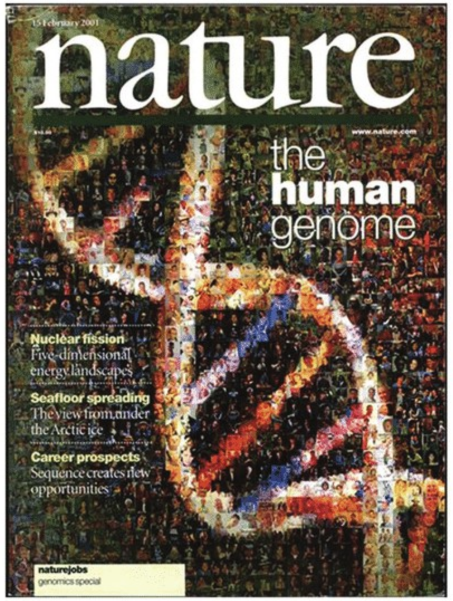
- PGS. TS. Lê Đình Thanh
- PGS. TS. Võ Thành Toàn
- ThS. Nguyễn Văn Tài
- ThS. Nguyễn Thị Duyên Anh
- BS.CK1 Trần Vũ Minh Tâm
- ThS. Cao Thị Hoài
- ThS. DS Lê Đặng Minh Anh
- CN. Trần Hoàng Uyên
- CN. Phạm Hiếu Đan
- CN. Vương Vũ Hoàng
- ThS. Thái Mỹ Ngân
- ThS. Lê Trí Viễn

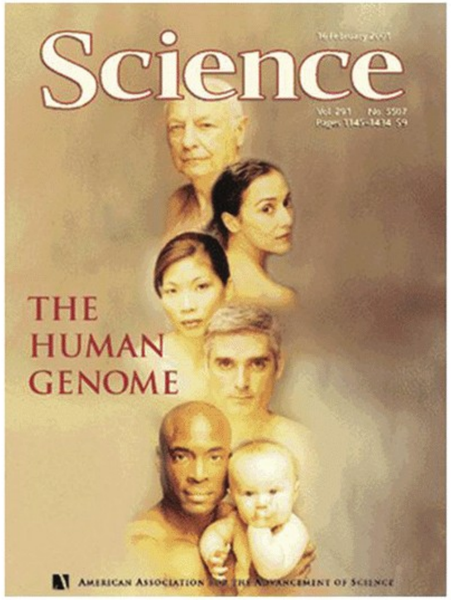
Giới thiệu

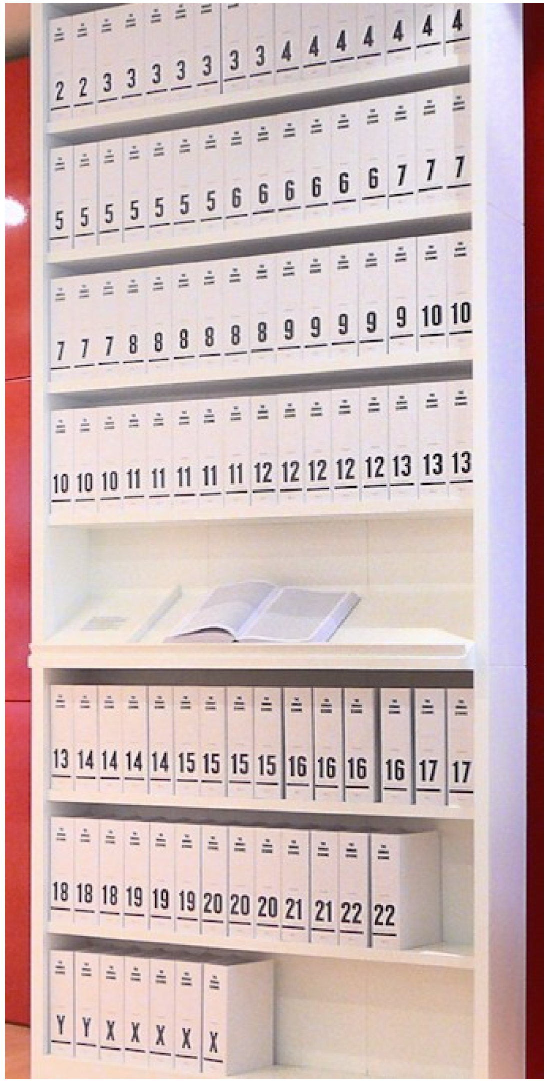
DỰ ÁN BỘ GEN NGƯỜI & GIẢI TRÌNH TỰ GEN THẾ HỆ MỚI

Dự án hệ gen người (1990-2003)

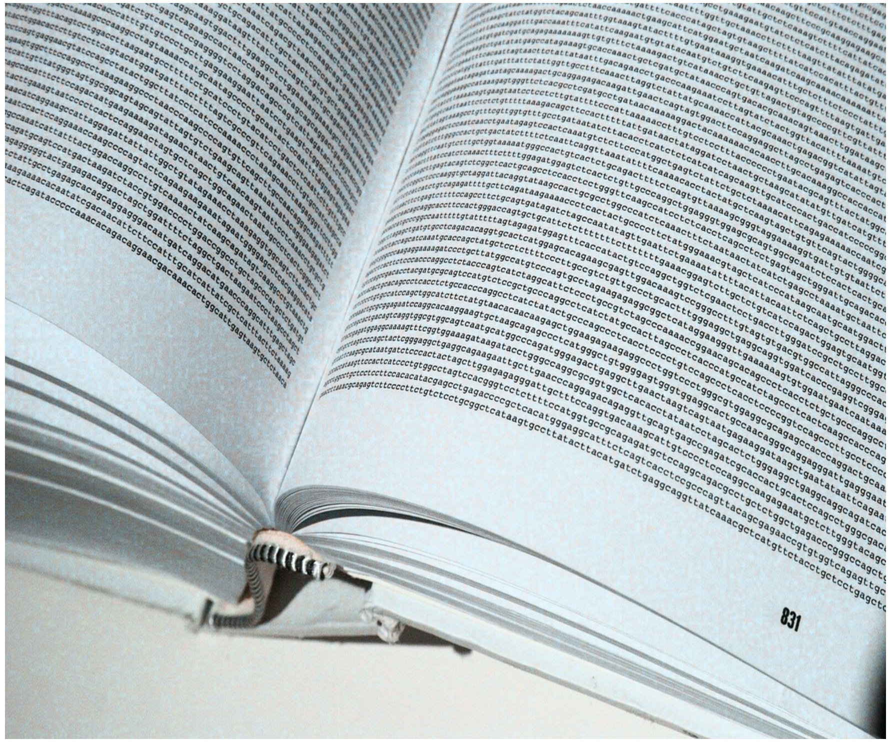








Năm 2000

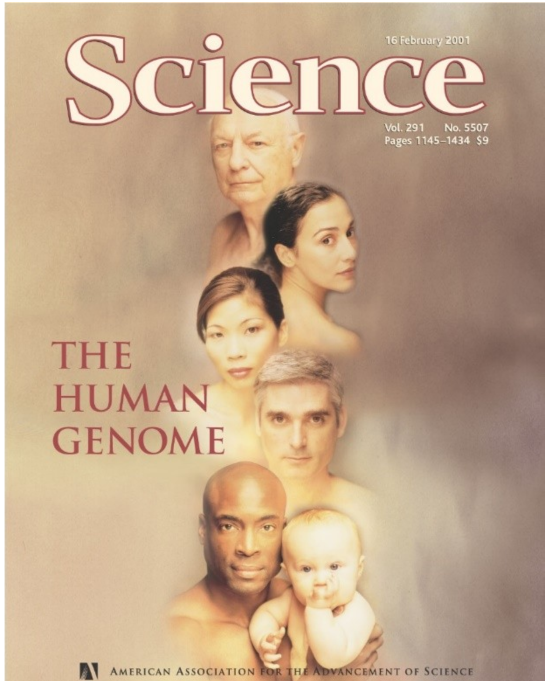


Dự án hệ gen người HGP (Oct 1990 - April 2003)

- Năm 2003, dự án hệ gen người HGP đã tạo ra một chuỗi trình tự gần 3 tỉ nucleotide chiếm hơn 90% bộ gen người.
- Đây là chuỗi gen hoàn chỉnh nhất có thể đạt được với công nghệ giải trình tự DNA thời điểm đầu những năm 2000.

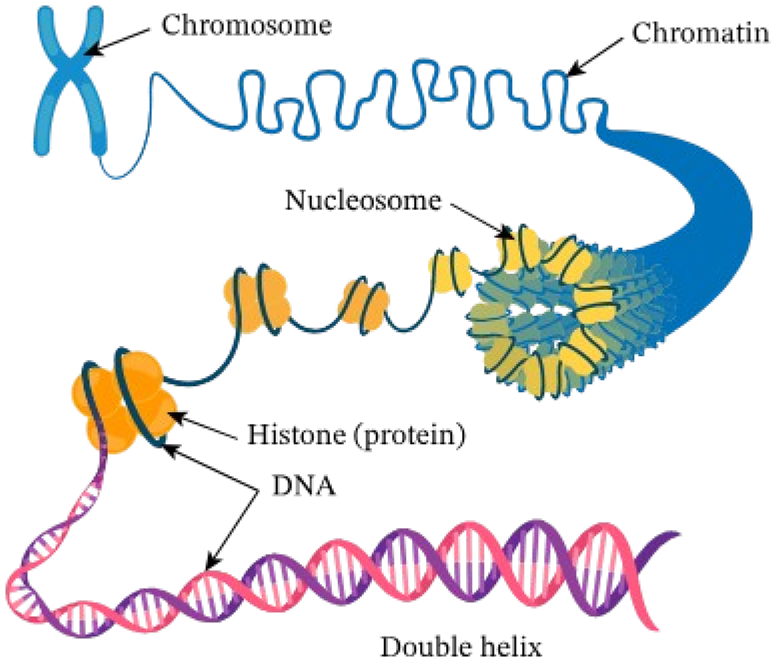
=> Sự phát triển của công nghệ giải trình tự thế hệ mới (NGS).

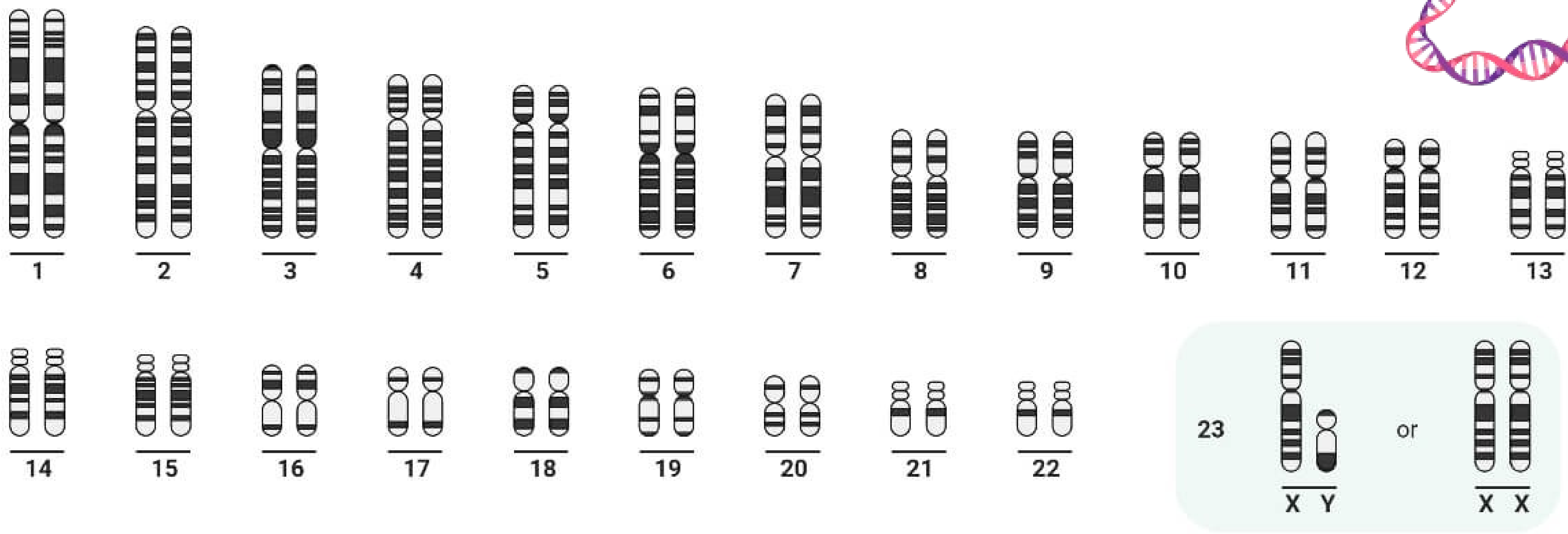
HGP Paper



Venter/Celera Paper

Human Karyotype





https://microbenotes.com/karyotype-karyotyping/

Hệ Gen người

```
48541 agcccttcaa agaaatgttc tcagcaggca tggagcccag gacttgctcc ctttgggtag
48601 agagccgggt tgaaggtgac tgaagtgaaa tgggacagta gaggcggggg gggtggtgag
48661 ttcctggagg tggggggtgt gggaacctgc tgtgtactga gatgcacccc tgccagttct
48721 gcctgaagat ttgaggcggg gggcaggggg gcggagtgaa gtcattttac tggtaagtaa
48781 ttttaaacct tttaatatta aagcaaacgt ggatatgtaa tgaatgaaat tcattctgga
48841 atgaaaaatt cacgtgatgt tgaaaaataa cacggggctt cagagaggac tttctggctg
48901 gcagcagact ccagattccc agggcccctg caccctcctc tgcccacagg gcaccttaat
48961 ggagaaggtg tgggaggaga gccaggccgg agtcagagca cactggtgac tccacatttg
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49081 ctgtccccgt ccagtttctg aggacaagag ccaccacctg tagcaaataa agacccagca
49141 accctttgac tcatctttgt gagtctctgg aatcagaggg tagccacatc gctgagaggt
49201 ggagtgaagc actcgggtga aaaggtacaa ggaagtcagg gacaggagtg tggggacatc
49261 acctagacaa tgacagagaa gaggggcaca gccgagtgag gggagagggg ccggcagtcc
49321 tacatcccct ggcctgaagc acgctccagg gcagaaggaa aaacactgtc tttggggtcc
49381 aagagacctg agttcaaatt ctggctccac cactgaccac ctgtgtaacc ttgaactgct
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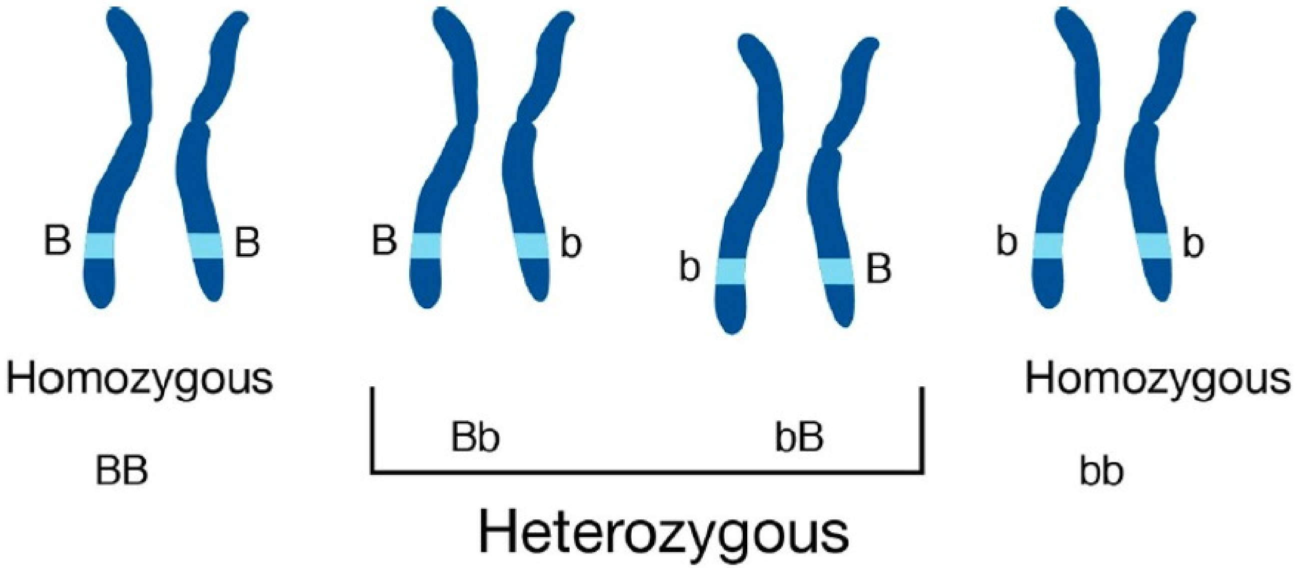
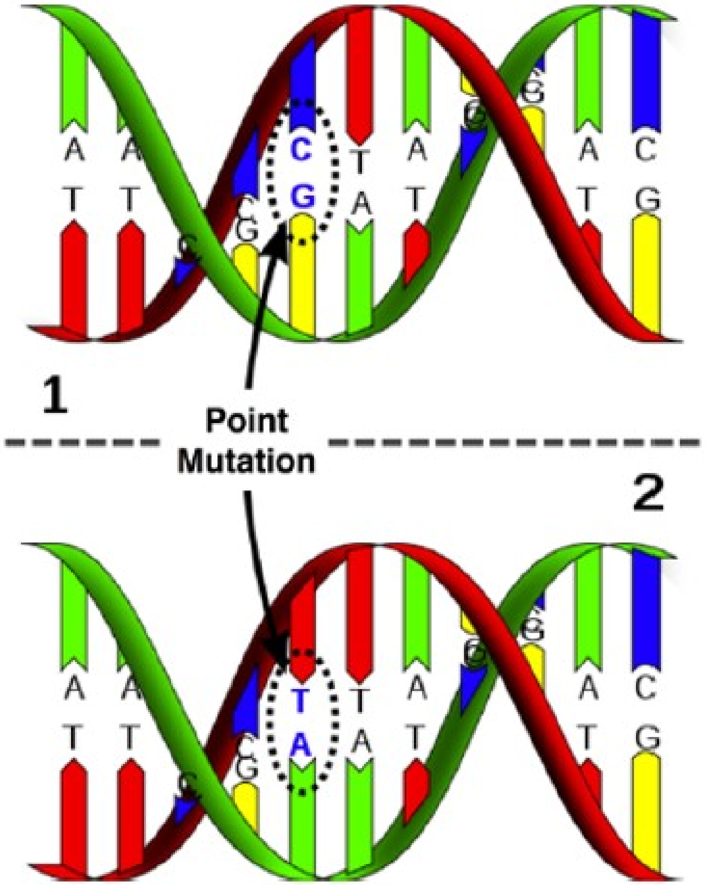
49501 gcccctgtga gaacgaaatg gtgcaagcac caaggagcct cagcaaaggt cgggcctgcc
49561 cccgcctggc caaacctttc ctcttcagga ggccacggca accgtagttt gacagaagag
49621 cagcaccttg atttaatgct tcccagcatg tgtccttgag caagtcacct aacctctctg
49681 ggctgcttcc tcattgggaa aatatggctg ccagtaaaac ctgccctgtc cacctcctgg
49741 ggcacttggc aaacagcaaa agagtccaaa tgtgcaggct gggccaggcg cagtggctca
49801 tgcctgtaat cccagcaatt taggaagcca aggtgggcgg atcacctgag gtcaggagtt
49861 tgagaccagc ctggccaaca tggtgaaacc ttgtctctac aaaaatacaa aaattagccg
49921 ggcatgatgg cgggtgcctg taatcccagt tactcgggag gctgaggcaa gagaatcgct
49981 tgaacccgga aggggaaggt tgcagtgagc caagattgtg ccactgcact ccagcctggg
50041 caacagagcg agactctgtc tcaaaaaaaa aaaaaaaaaa aaacaatgca gagctggctg
50101 tgtaaaaaac ctgttccact gcagggccca gtgtccacca ggctggggtg caggcctatg
50161 gggtgggggc ccagcatcag cctctcagca gccctgggag gcggggcgca tcccgtgccc
50221 ctcgtggtct ggatgtgttc tagcccaagt cctaggttac acctgccgtc gcctggcctc
50281 tcaggagagg cccagggtga ggaggagcat ggtaaaggtg aagctgattg ggaagtcagc
50341 tgttgggaaa gcaactcctt gcacattgga ggaaccgaga aagactgacc ccgaggacag
50401 cagccagcat ggcccttcct gggagcccat gttgggggat tcctgctgca gccaaggctc
50461 agcccttgtg gtcgcaggtg ctggtcctgg cctcttcccc tcccatgcag gagcacagga
50521 gagatggctt ctgaggacct gttgcagctg tggccctggg aatagatttg ccagggagct
50581 ttaaagcagc tgagtgtgtc atccagctaa gcctggggaa ggagcttggc tcaggtcctg
50641 acaggtgtga cagggatggg gactgggaag taagagatga aaccctggct ggaggctgtg
50701 agcttccaca gccagcgctg gacagggagg gtccagatat acccactagt gccctcacca
```

https://www.ncbi.nlm.nih.gov/nuccore/806904736

Các loại biến thể trên hệ gen

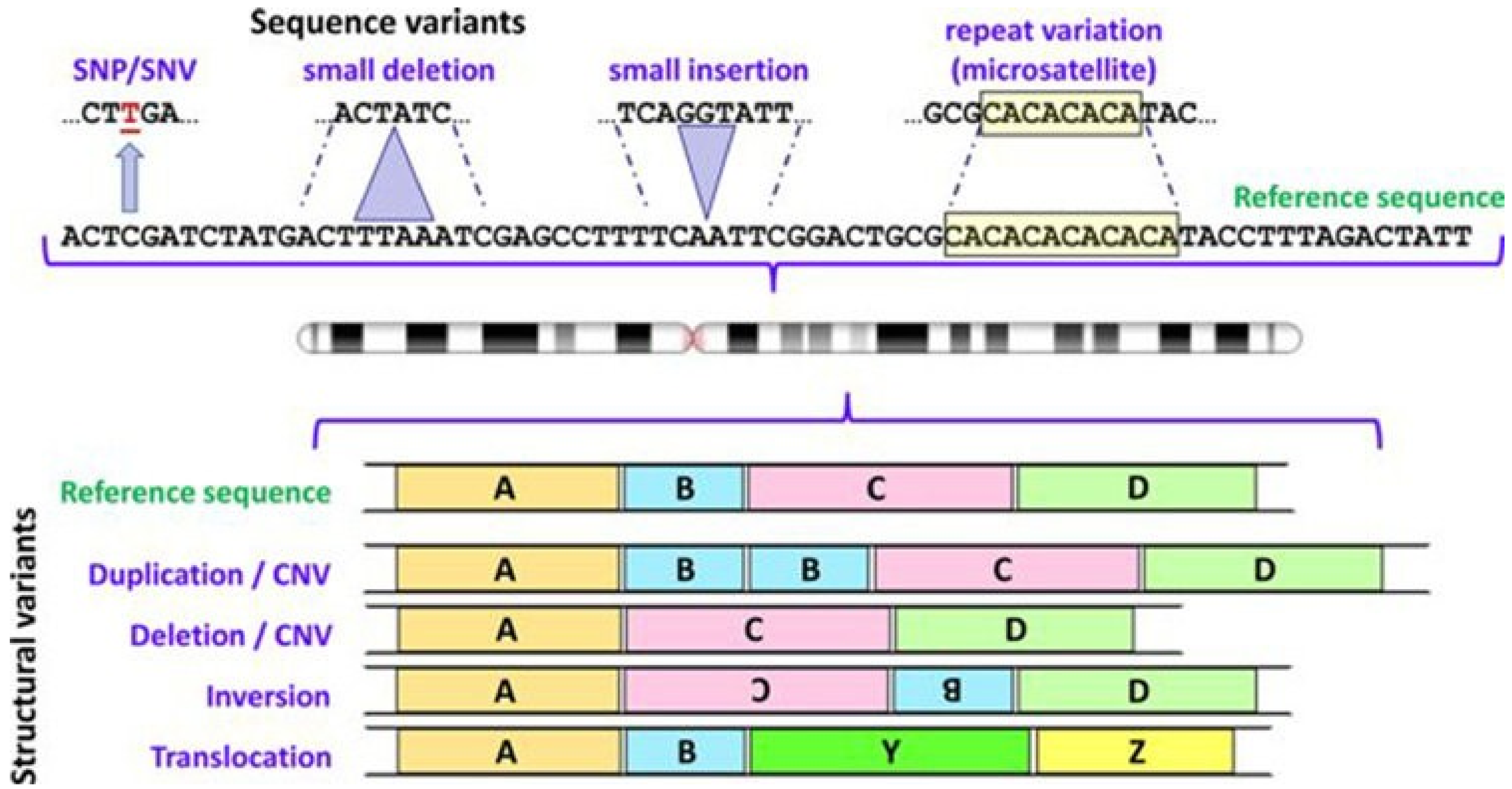
- Hệ gen giữa hai người giống nhau > 99%
- Mỗi người có khoảng 5 triệu biến thể, trong đó có 3 đến 4 triệu biến thể một nucleotide
- Hệ gen người là hệ lưỡng bội





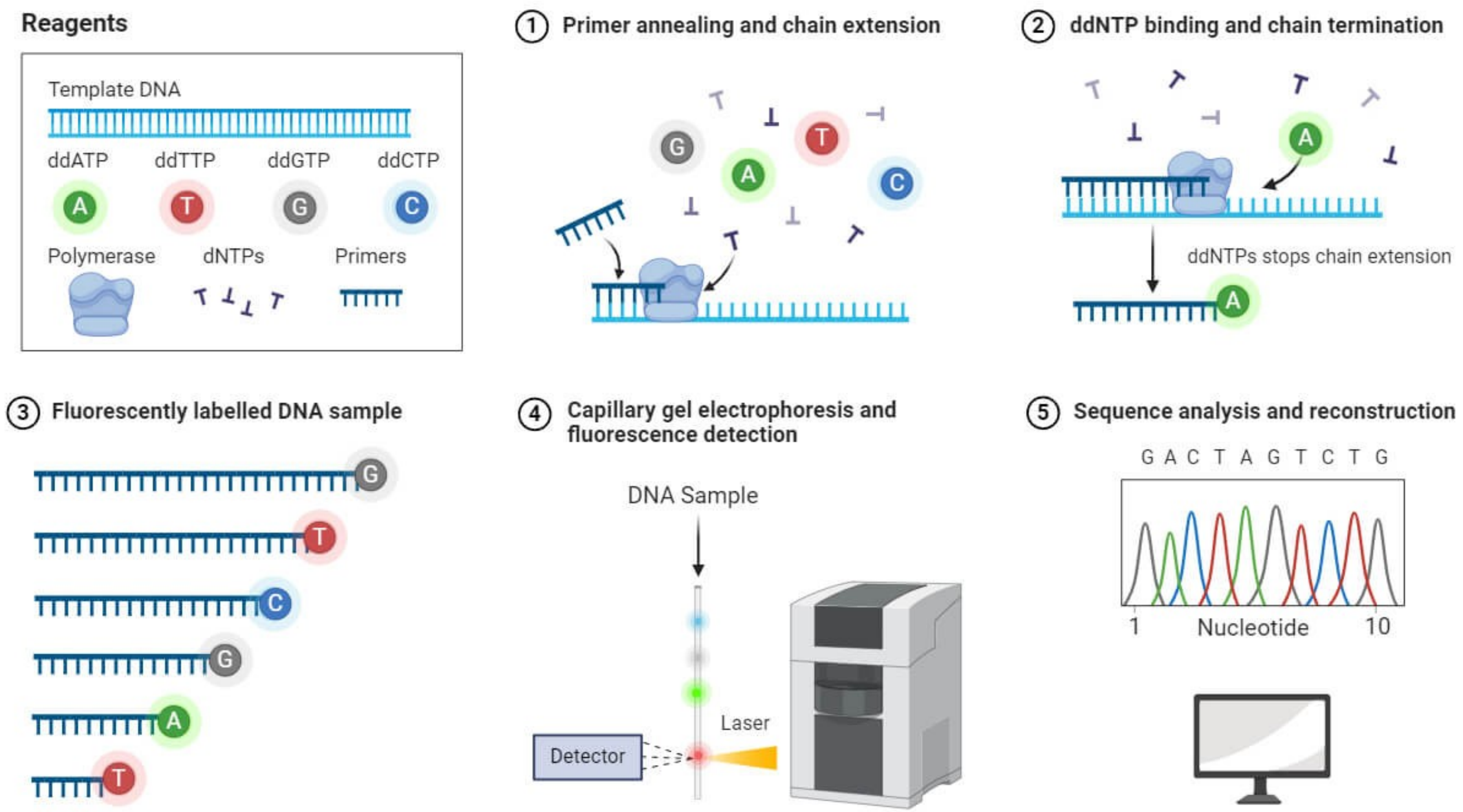
https://rosalind.info/media/point_mutation.png, https://en.wikipedia.org/wiki/Zygosity

Các loại biến thể trên hệ gen



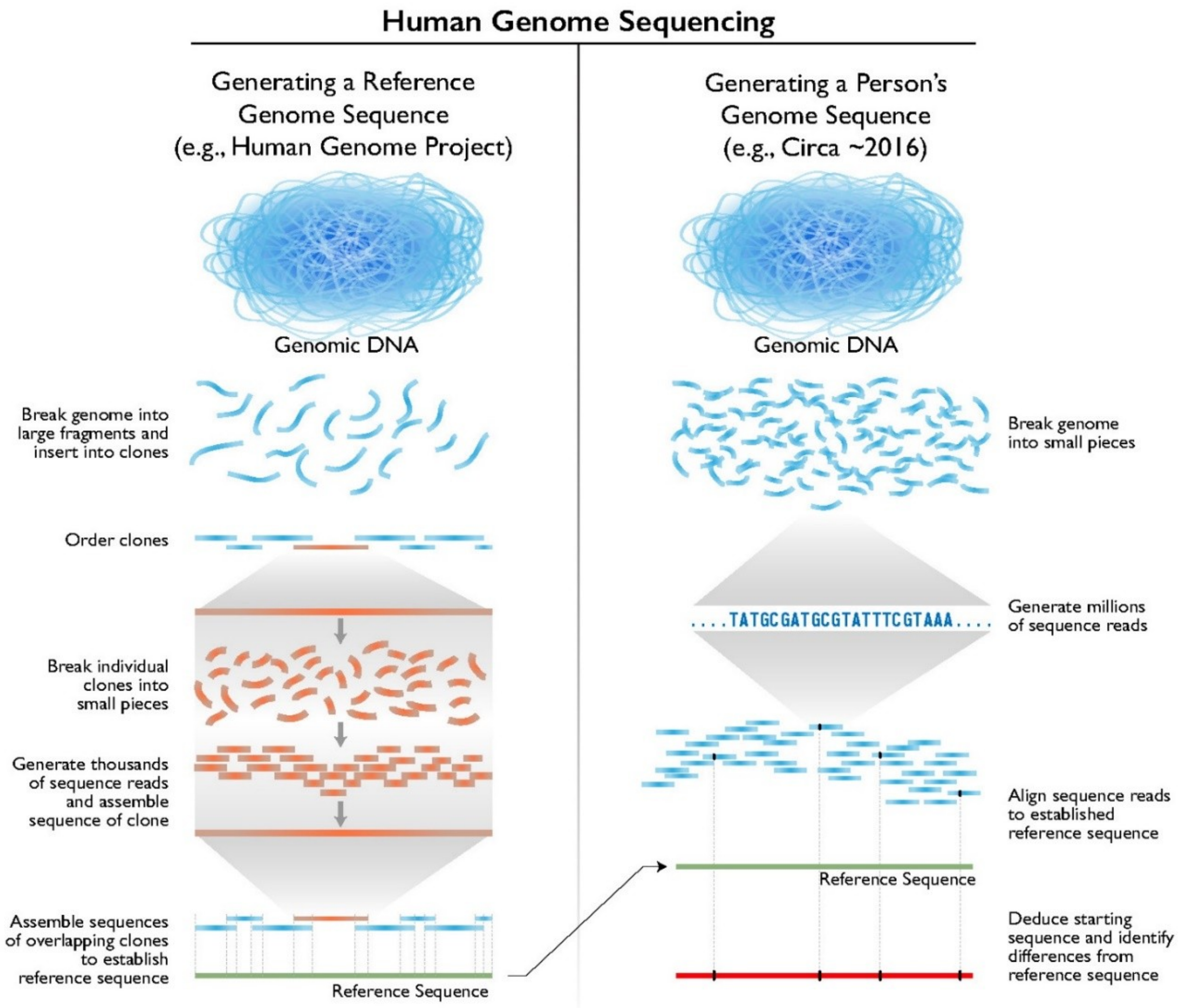
10.1042/EBC20170053

First Generation Sequencing (Sanger Sequencing)



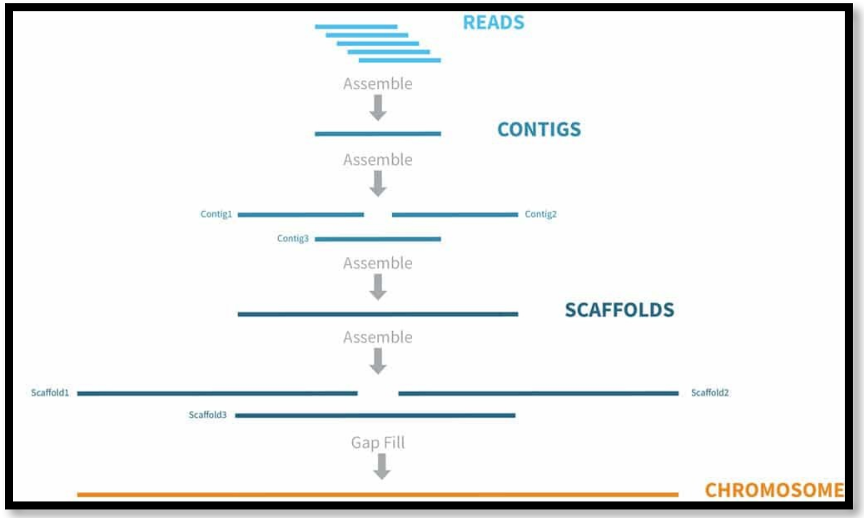
- https://www.youtube.com/watch?v=KTstRrDTmWI
- https://www.youtube.com/watch?v=X9566yI2cBo

Giải trình tự gen thế hệ mới (NGS)

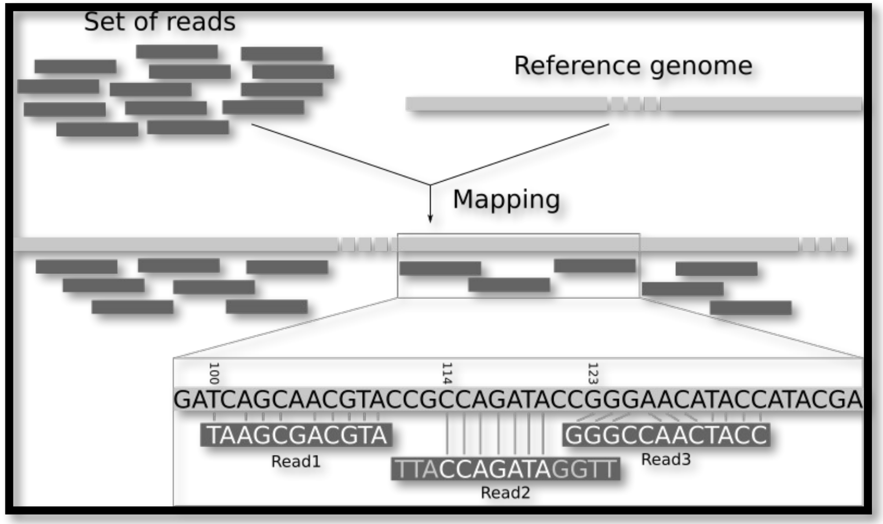


https://www.genome.gov/about-genomics/fact-sheets/Sequencing-Human-Genome-cost

De novo assembly



Mapping to reference



Ví dụ nhỏ: N = 5 -> 5^2
Ví dụ ở vi khuẩn giải 100X: N = (3,500,000 x 100)/100 = 3,500,000 PE100 → $(3,500,000)^2$
Ví dụ ở người giải 30X: N = (3,000,000,000 x 30)/150 = 600,000,000 PE150 → $(600,000,000)^2$

1) ACTCAGAT
2) ATGTACTCA
3) CAGATTTGC
4) GATCAGTCA
5) CATCAGTTT

De Novo Assembly	1	2	3	4	5
1		x	x	x	x
2			x	x	x
3				x	x
4					x
5					

Mapping	Reference Genome
1	x
2	x
3	x
4	x
5	x

De Novo Assembly

```
2)  ATGTACTCA
1)      ACTCAGAT
3)         CAGATTTGC
→   ATGTACTCAGATTTGC
```

Mapping to Reference Genome

```
Ref  ATGTACTCAGATTTGC
         ACTCAGAT
     ATGTACTCA
            CAGATTTGC
```

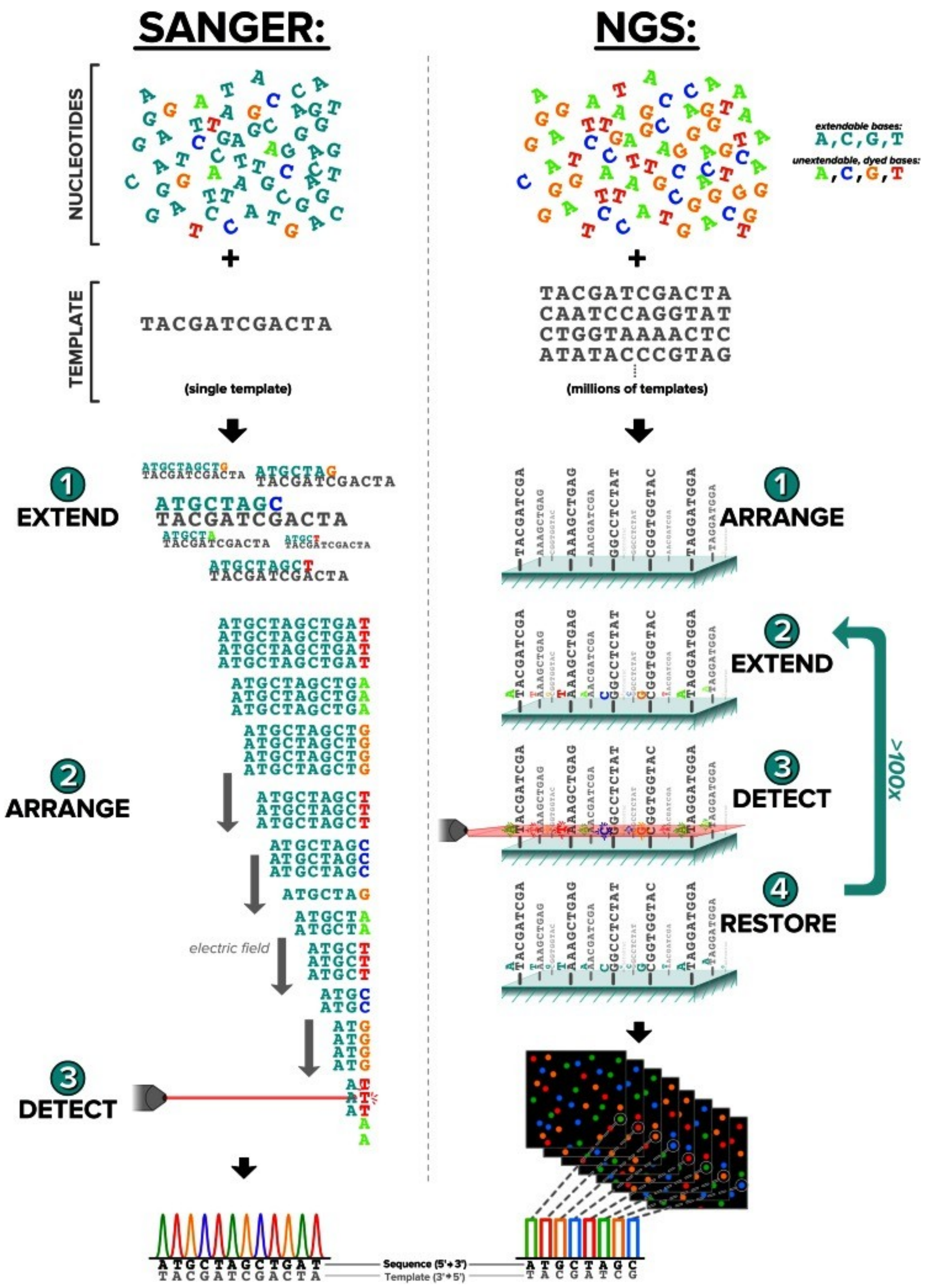
Giải trình tự gen thế hệ mới (NGS): Có hệ gen tham chiếu

De novo assembly

Mapping to reference

Giải trình tự gen thế hệ mới (NGS): giải trình tự song song

Read1:CTCGAATACG

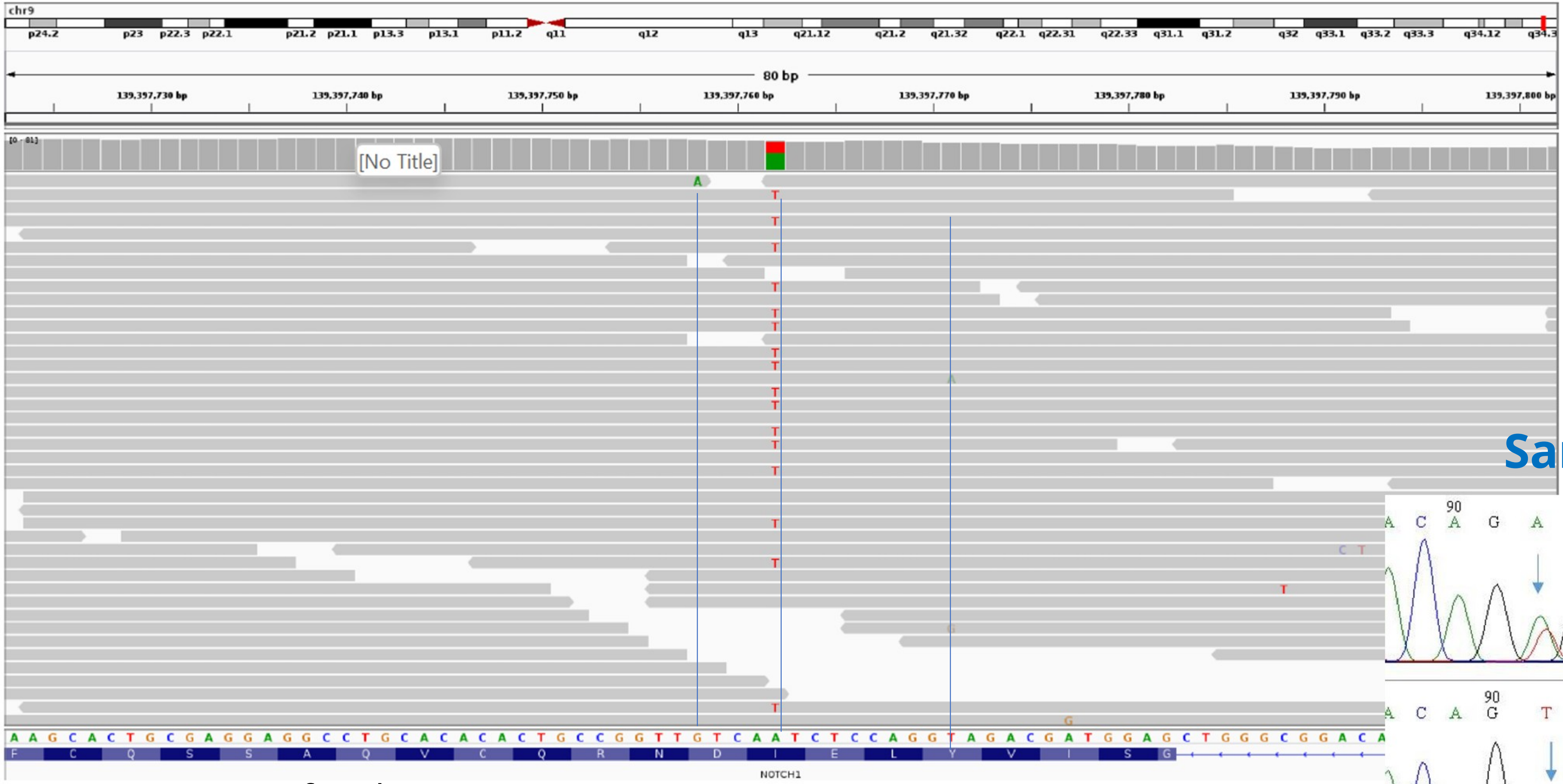


Giải trình tự gen thế hệ mới (NGS): chuẩn bị thư viện có kit thương mại

Read1:CTCGAATACG
Read2:CTCGAATACG
Read3:CTCGAATACG
Read4:CTCGAATACG
Read5:CGCGAATACG
Read6:CGCGAATACG
Read7:CGCGACTACG
Read8:CGCGAATACG

10.1007/s40142-015-0076-8

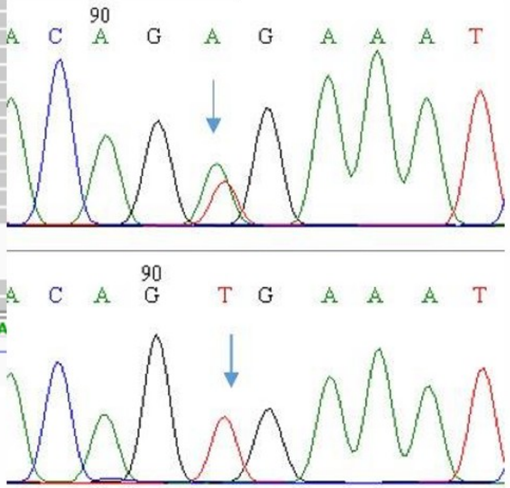
Trực quan hóa dữ liệu giải trình tự DNA bằng NGS



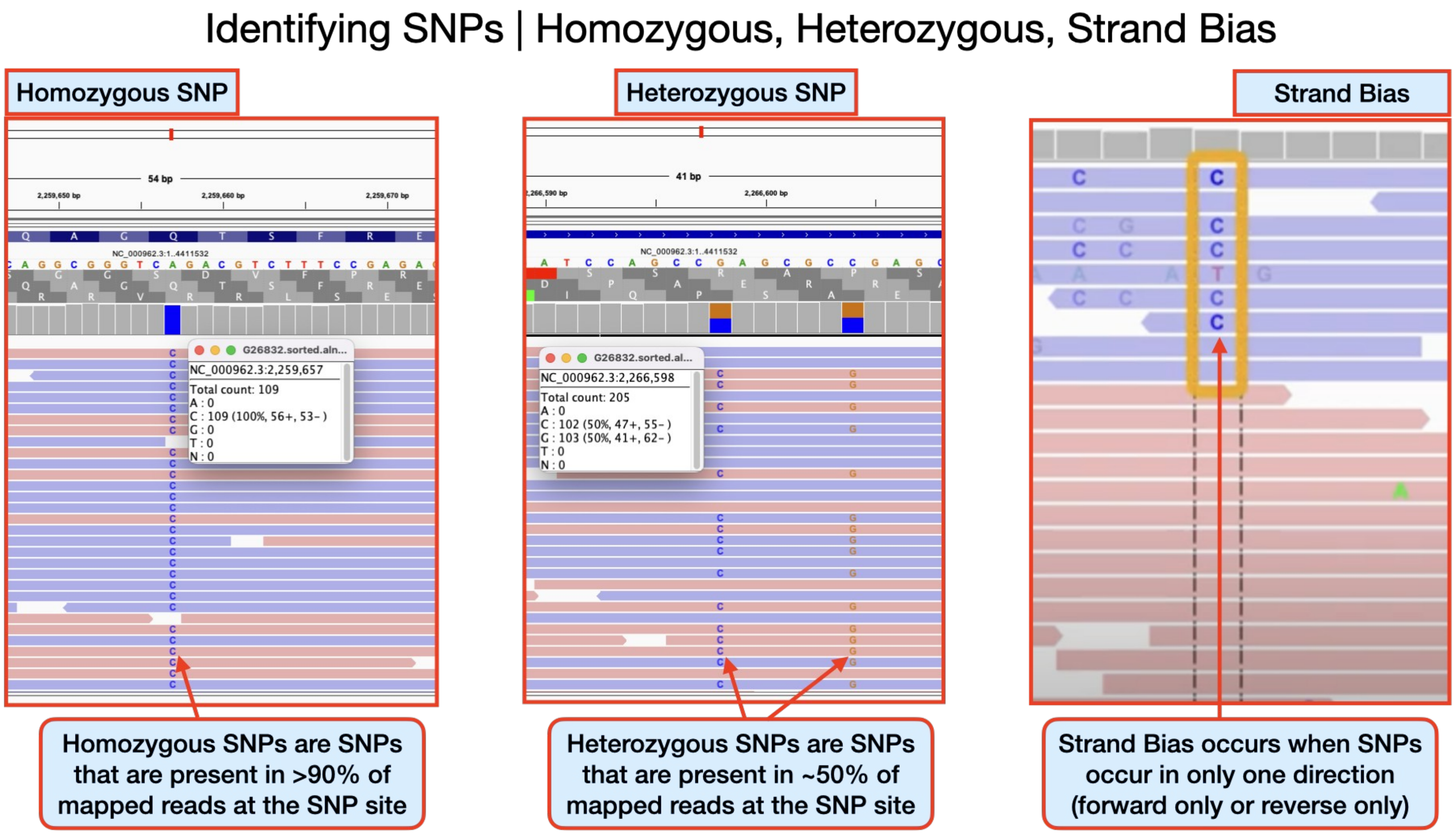
Ref 10th G: 99G, 1A (100X)

Ref 14th A: 55A, 45T (100X)

Sanger

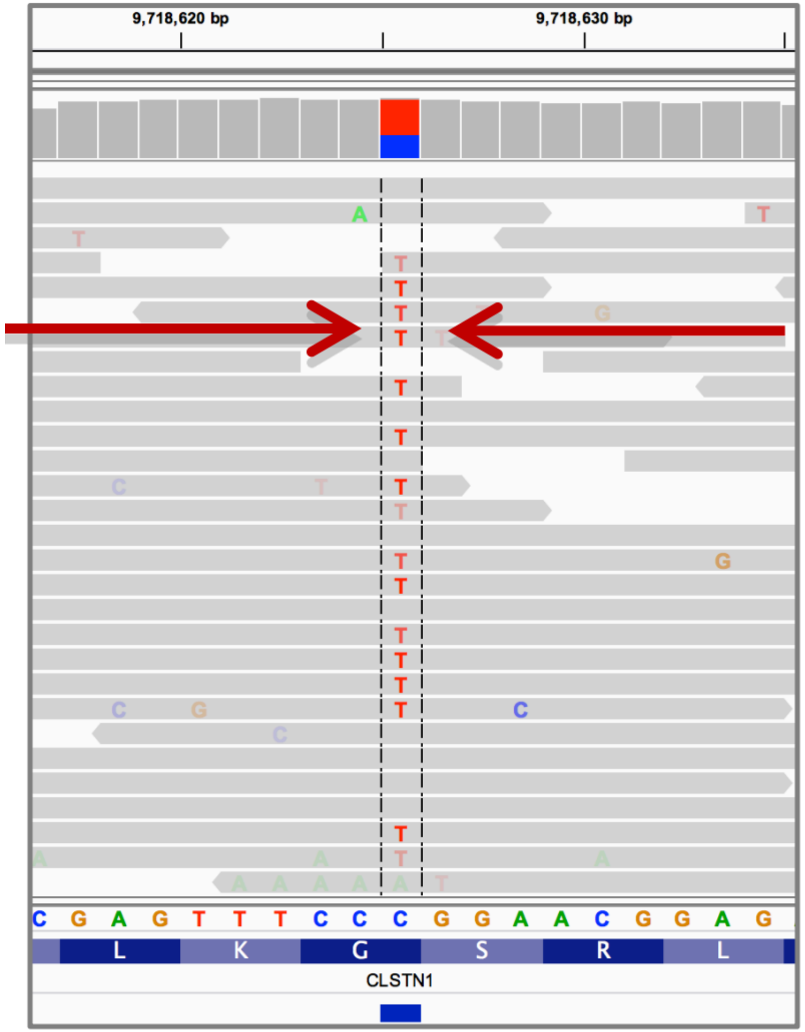


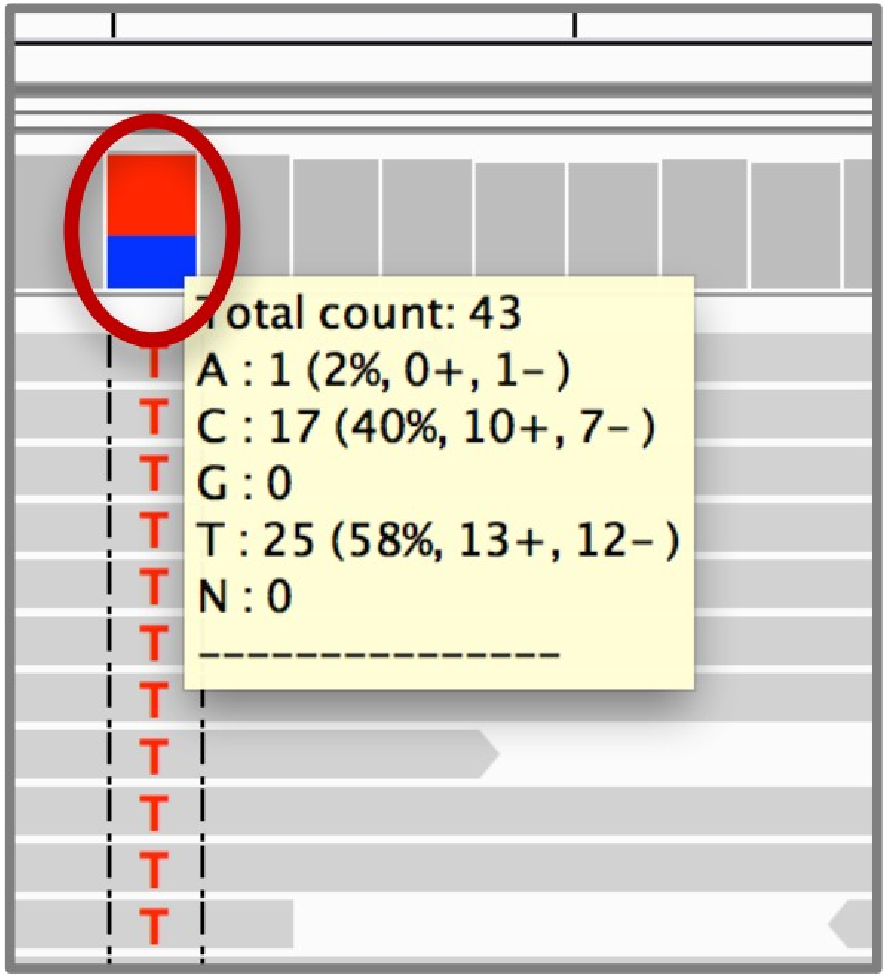
Đồng hợp tử hay dị hợp tử?



https://avantonder.github.io/Ghana_course/materials/05-short_read_mapping/5.1-short_read_mapping.html

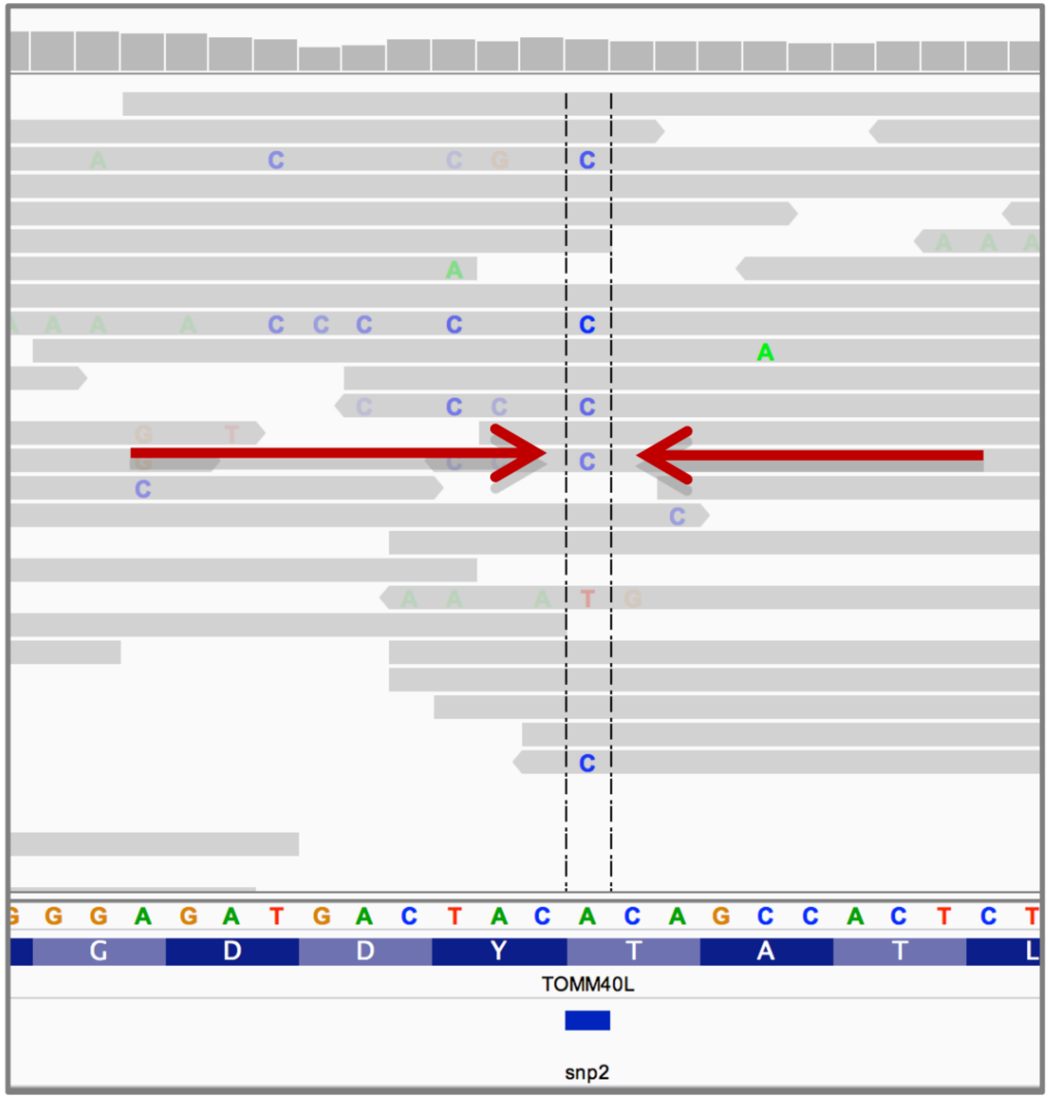
Đồng hợp tử hay dị hợp tử?

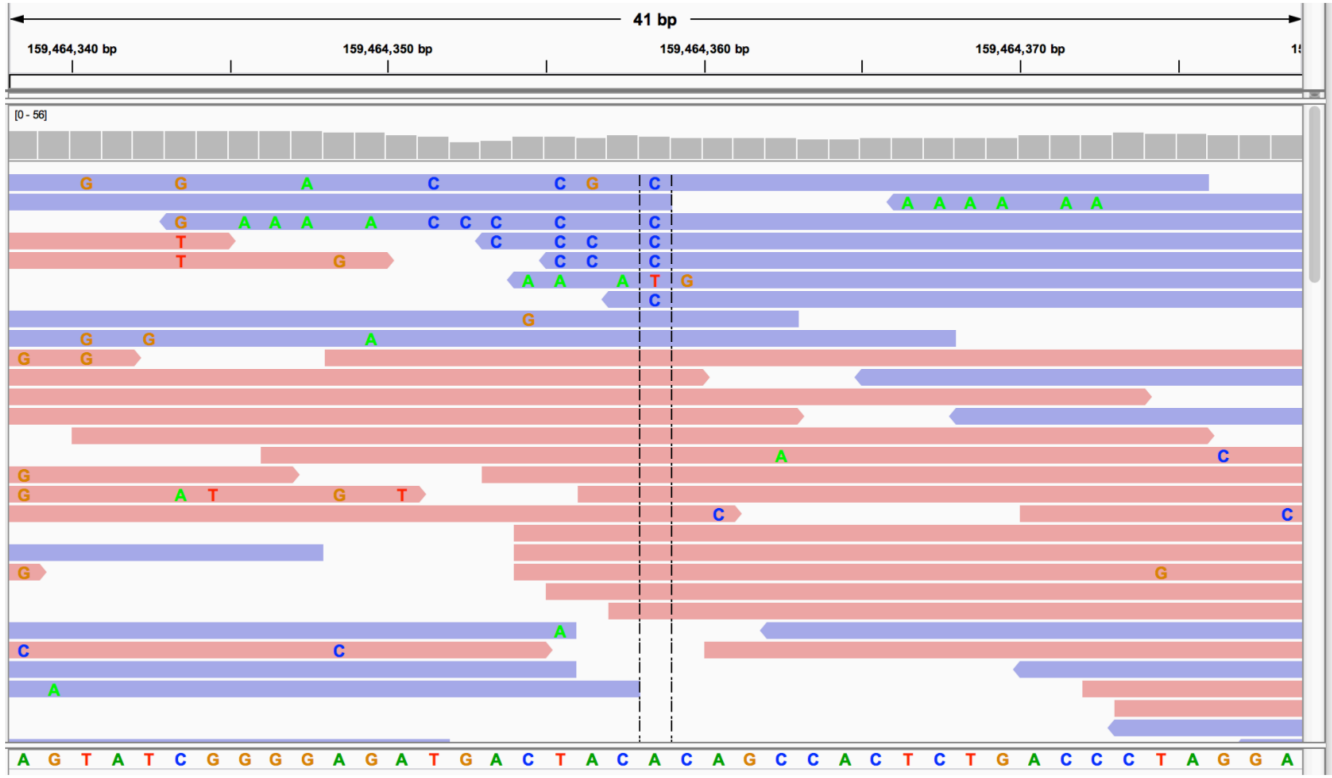




https://datasets.genepattern.org/data/workshops/240411-PSTP/3_SNPs.pdf

Đồng hợp tử hay dị hợp tử?



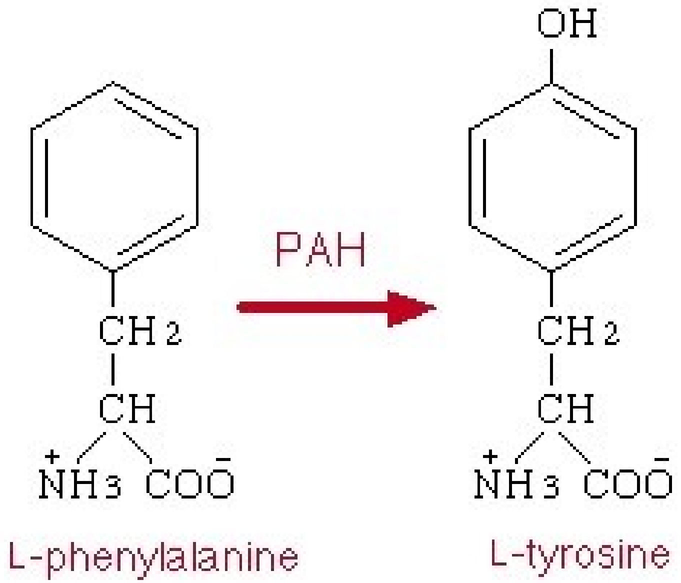


https://datasets.genepattern.org/data/workshops/240411-PSTP/3_SNPs.pdf

Ví dụ mối quan hệ của
BIẾN THỂ GEN và BỆNH DI TRUYỀN

Phenylketonuria (PKU): Rối loạn chuyển hóa di truyền

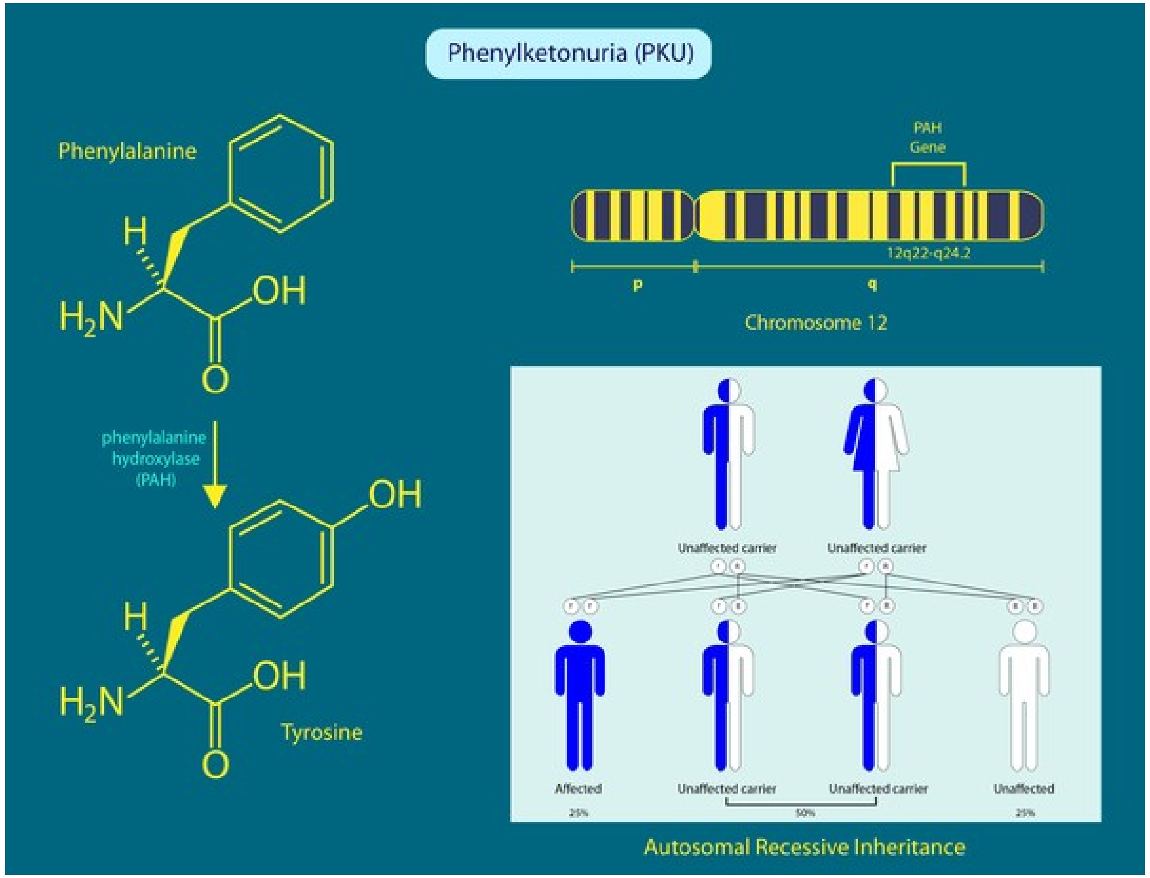
- Nguyên nhân do thiếu hụt enzyme phenylalanine hydroxylase.
- Sự mất enzyme này dẫn đến suy giảm trí tuệ, tổn thương cơ quan, tư thế bất thường.
- Tần suất xảy ra PKU khác nhau giữa các nhóm dân tộc và các vùng địa lý trên toàn thế giới. Ở Hoa Kỳ, PKU xảy ra ở 1 trong 25.000 trẻ sơ sinh.
- Hầu hết các trường hợp PKU được phát hiện ngay sau sinh bằng sàng lọc sơ sinh và điều trị được bắt đầu ngay lập tức.



The enzyme phenylalanine hydroxylase converts the amino acid phenylalanine to tyrosine.

https://www.ncbi.nlm.nih.gov/books/NBK22253/

Phenylketonuria (PKU): Rối loạn chuyển hóa di truyền



https://medlineplus.gov/genetics/condition/phenylketonuria/

Tên khác của PKU

- Folling disease
- Folling's disease
- PAH deficiency
- Phenylalanine hydroxylase deficiency
- Phenylalanine hydroxylase deficiency disease

Trình tự gen PAH ở người - Homo sapiens (5053)

Gene (Nucleotide)

NT seq	1359 nt NT seq
	atgtccactgcggtcctggaaaacccaggcttgggcaggaaactctctgactttggacag gaaacaagctatattgaagacaactgcaatcaaaatggtgccatatcactgatcttctca ctcaaagaagaagttggtgcattggccaaagtattgcgcttatttgaggagaatgatgta aacctgacccacattgaatctagaccttctcgtttaaagaaagatgagtatgaatttttc acccatttggataaacgtagcctgcctgctctgacaaacatcatcaagatcttgaggcat gacattggtgccactgtccatgagctttcacgagataagaagaaagacacagtgccctgg ttcccaagaaccattcaagagctggacagatttgccaatcagattctcagctatggagcg gaactggatgctgaccaccctggttttaaagatcctgtgtaccgtgcaagacggaagcag tttgctgacattgcctacaactaccgccatgggcagcccatccctcgagtggaatacatg gaggaagaaaagaaaacatggggcacagtgttcaagactctgaagtccttgtataaaacc catgcttgctatgagtacaatcacatttttccacttcttgaaaagtactgtggcttccat gaagataacattccccagctggaagacgtttctcagttcctgcagacttgcactggtttc cgcctccgacctgtggctggcctgctttcctctcgggatttcttgggtggcctggccttc cgagtcttccactgcacacagtacatcagacatggatccaagcccatgtataccccccgaa cctgacatctgccatgagctgttgggacatgtgcccttgttttcagatcgcagctttgcc cagttttcccaggaaattggccttgcctctctgggtgcacctgatgaatacattgaaaag ctcgccacaatttactggtttactgtggagtttgggctctgcaaacaaggagactccata aaggcatatggtgctgggctcctgtcatcctttggtgaattacagtactgcttatcagag aagccaaagcttctcccccctggagctggagaagacagccatccaaaattacactgtcacg gagttccagcccctctattacgtggcagagagttttaatgatgccaaggagaaagtaagg aactttgctgccacaatacctcggcccttctcagttcgctacgacccatacacccaaagg attgaggtcttggacaatacccagcagcttaagattttggctgattccattaacagtgaa attggaatcctttgcagtgccctccagaaaataaagtaa

Protein (Amino Acid)

AA seq	452 aa AA seq DB search
	MSTAVLENPGLGRKLSDFGQETSYIEDNCNQNGAISLIFSLKEEVGALAKVLRLFEENDV NLTHIESRPSRLKKDEYEFFTHLDKRSLPALTNIIKILRHDIGATVHELSRDKKKDTVPW FPRTIQELDRFANQILSYGAELDADHPGFKDPVYRARRKQFADIAYNYRHGQPIPRVEYM EEEKKTWGTVFKTLKSLYKTHACYEYNHIFPLLEKYCGFHEDNIPQLEDVSQFLQTCTGF RLRPVAGLLSSRDFLGGLAFRVFHCTQYIRHGSKPMYTPEPDICHELLGHVPLFSDRSFA QFSQEIGLASLGAPDEYIEKLATIYWFTVEFGLCKQGDSIKAYGAGLLSSFGELQYCLSE KPKLLPLELEKTAIQNYTVTEFQPLYYVAESFNDAKEKVRNFAATIPRPFSVRYDPYTQR IEVLDNTQQLKILADSINSEIGILCSALQKIK

https://www.genome.jp/entry/T01001:5053

Gen PAH (Năm 2009)

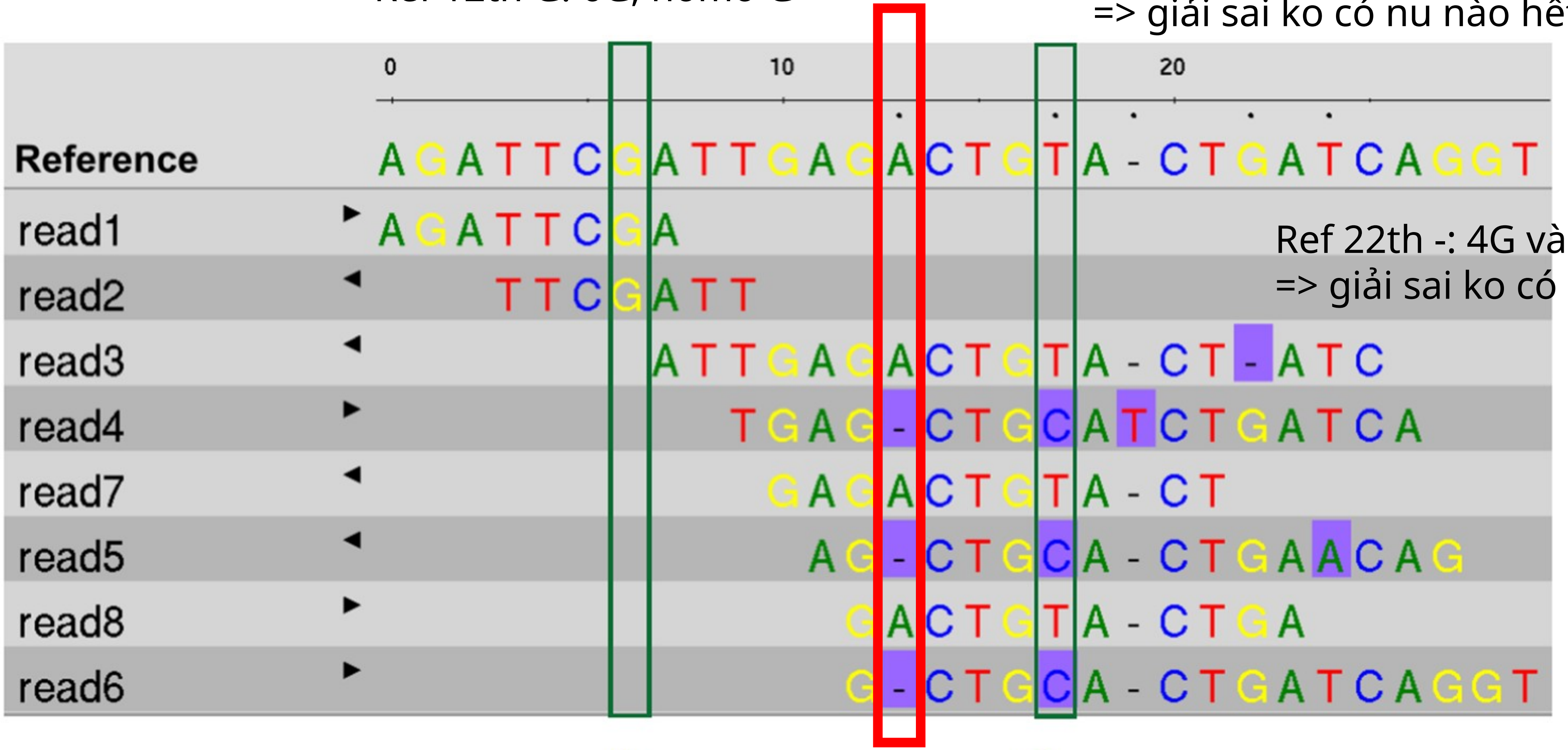
https://www.ncbi.nlm.nih.gov/gene/5053

Variants in PAH

Ref 12th G: 6G, homo G

Ref 19th -: T và 5-, homo -
=> giải sai ko có nu nào hết!

Ref 22th -: 4G và 1-, homo G
=> giải sai ko có bị mất nu!



2x

6x

Ref 13rd A: 3A và 3-, het A/-

Ref 17th T: 3T và 3C, het T/C

Cơ sở dữ liệu ClinVar (Năm 2013) cho gen PAH

Classification type

- ☐ Germline (208)
- ☐ Somatic (0)

Germline classification

- ☐ Conflicting classifications (2)
- ☐ Benign (10)
- ☐ Likely benign (21)
- ☐ Uncertain significance (41)
- ☐ Likely pathogenic (40)
- ☐ Pathogenic (39)

Types of conflicts

- ☐ P/LP vs LB/B (0)
- ☐ P/LP vs VUS (0)
- ☐ VUS vs LB/B (2)

Molecular consequence

- ☐ Frameshift (20)
- ☐ Missense (96)
- ☐ Nonsense (7)
- ☐ Splice site (9)
- ☐ ncRNA (0)
- ☐ Near gene (0)
- ☐ UTR (27)

Variation type

- ☐ Deletion (39)

Links from Gene

Display options ▾ Sort by Relevance ▾ Download ▾ Items: 1 to 100 of 209

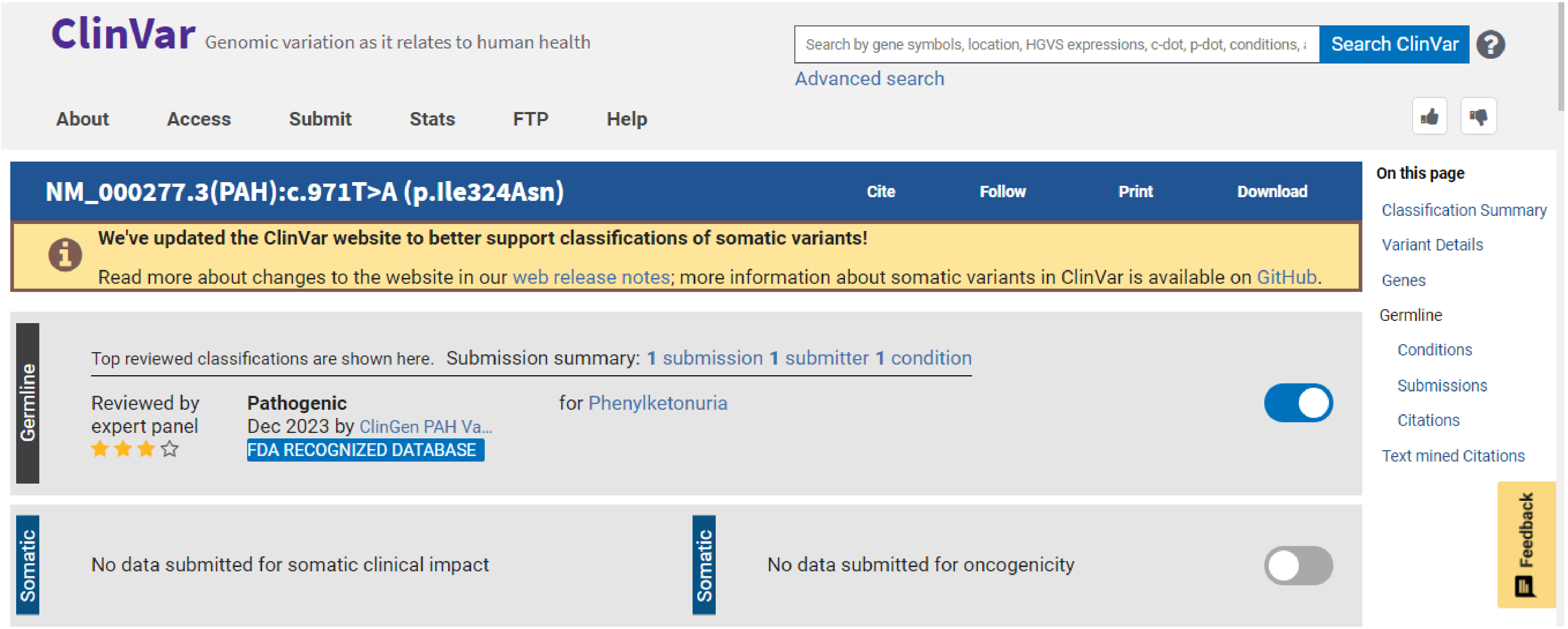
<< First < Prev Page 1 of 3 Next > Last >>

Variation	Gene (Protein Change)	Type (Consequence)	Condition	Classification, Review status
NM_004316.4(ASCL1):c.51G>T (p.Gln17His)	ASCL1, PAH (Q17H)	Single nucleotide variant (missense variant +1 more)	not specified	G Uncertain significance ★
NC_000012.11:g.(?_103232953)_(103240749_?)del	PAH	Deletion	Phenylketonuria	G Pathogenic ★
NC_000012.11:g.(?_103288493)_(103310908_?)del	PAH	Deletion	Phenylketonuria	G Pathogenic ★
NC_000012.11:g.(?_103248894)_(103249131_?)del	PAH	Deletion	Phenylketonuria	G Pathogenic ★
NC_000012.12:g.(?_102894715)_(102894938_?)del	PAH	Deletion	Phenylketonuria	G Pathogenic ★
NC_000012.11:g.(?_103306549)_(103306696_?)del	PAH	Deletion	Phenylketonuria	G Pathogenic ★
NM_000277.3(PAH):c.1179_1180del (p.Asn393fs)	PAH (N393fs)	Deletion (frameshift variant)	Phenylketonuria	G Likely pathogenic ★

https://www.ncbi.nlm.nih.gov/clinvar?LinkName=gene_clinvar&from_uid=5053

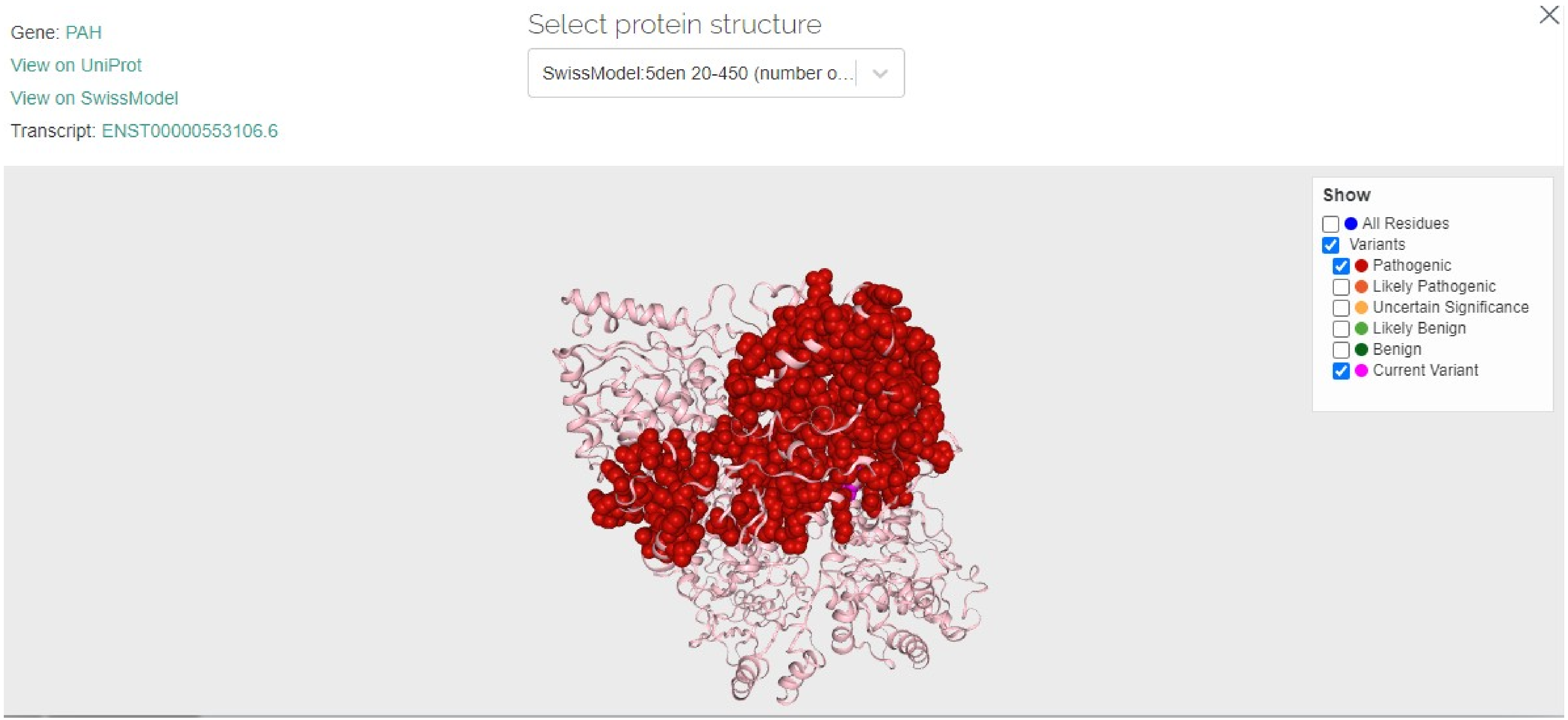
Biến thể gây bệnh - Pathogenic variant in PAH

NM_000277.3(PAH):c.971T>A (p.Ile324Asn)



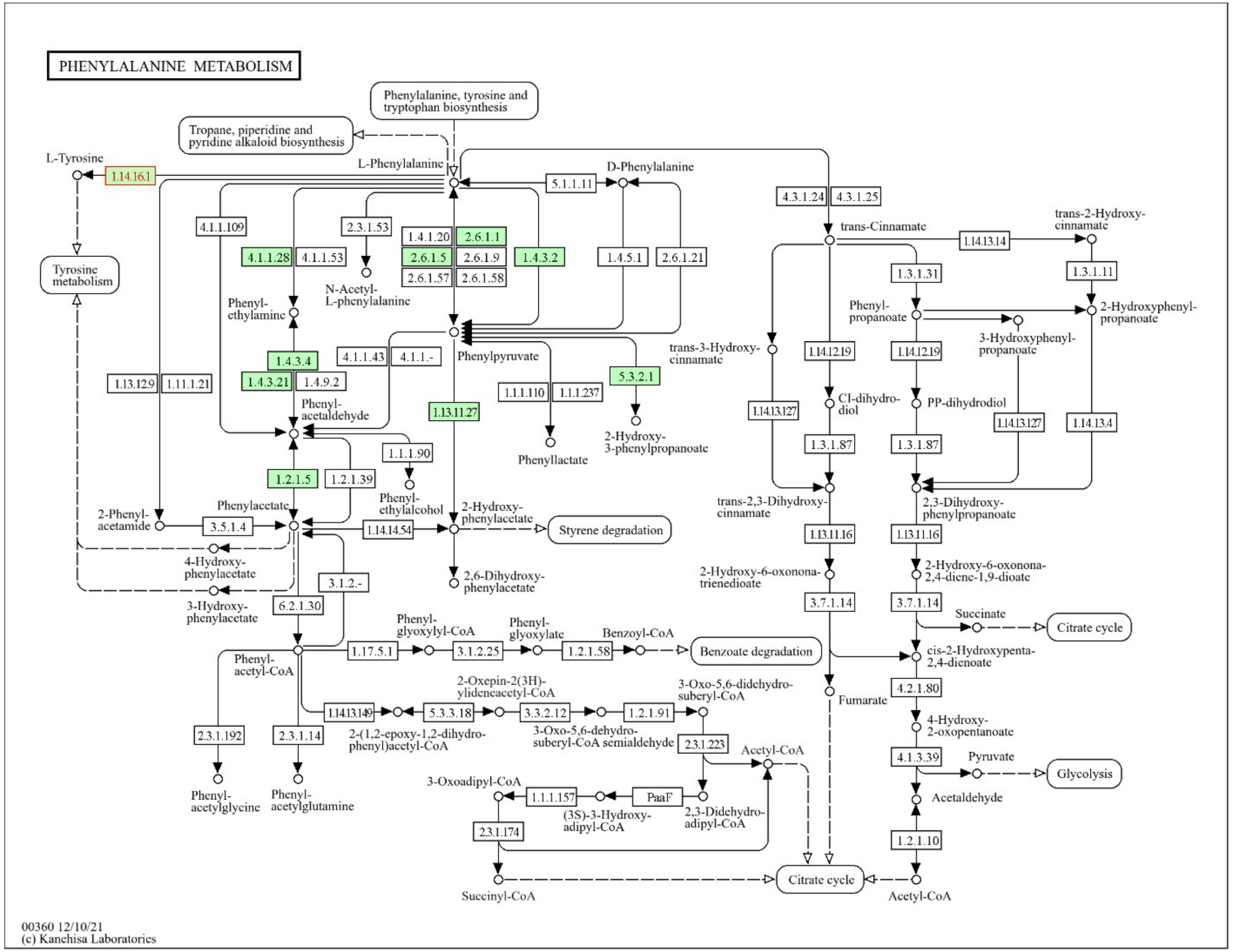
https://www.ncbi.nlm.nih.gov/clinvar/variation/2682170/

Các biến thể gây bệnh trên gen PAH



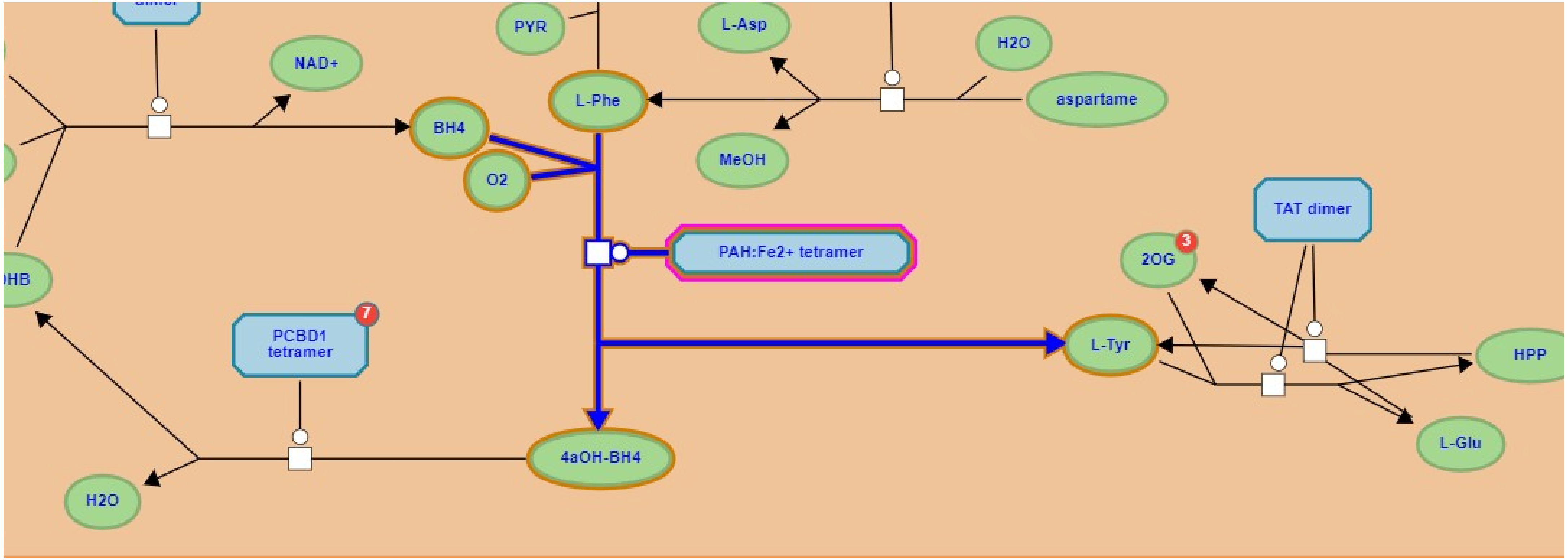
https://varsome.com/variant/hg38/chr12%3A102844430%3AA%3AT?

PAH: chuyển hóa Phenylalanine thành Tyrosine



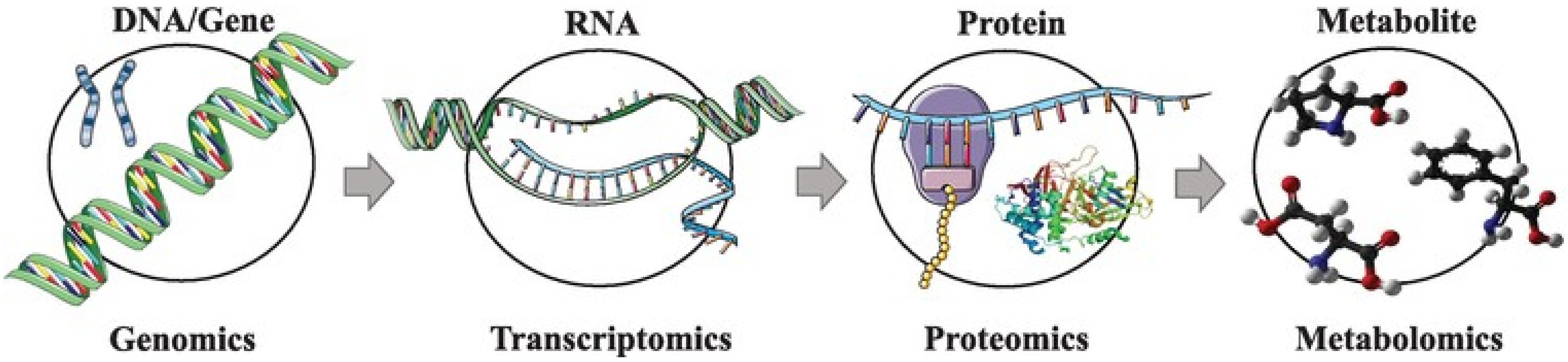
https://www.genome.jp/pathway/hsa00360+5053

PAH: chuyển hóa Phenylalanine thành Tyrosine



https://reactome.org/PathwayBrowser/#/R-HSA-8963691&SEL=R-HSA-71118&PATH=R-HSA-1430728,R-HSA-71291&FLG=UniProt:P00439

Mối liên kết: Biến thể gen và bệnh di truyền



PAH gene
Ref …ATCGAT…
P1 …AACGAT…

NM_000277.3(PAH):c.971T>A

PAH mRNA
Ref …AUCGAU…
P1 …AACGAU…

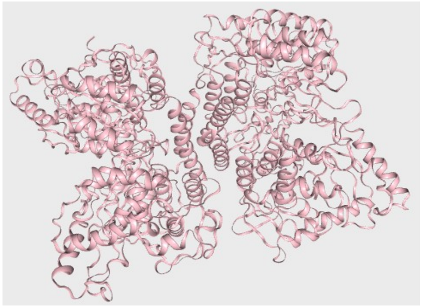
NM_000277.3(PAH):c.971T>A

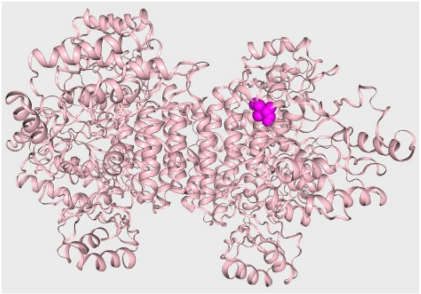
PAH protein
Ref …Ile-Asp…
P1 …Asn-Asp…

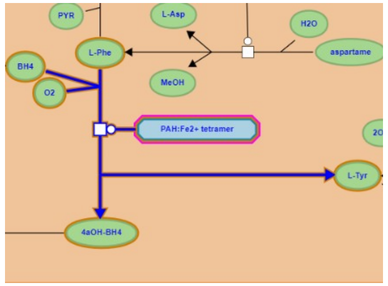
NM_000277.3(PAH):p.Ile324Asn

PAH
Ref Phe → Tyr

PAH
P1 Phe ~~→~~ Tyr

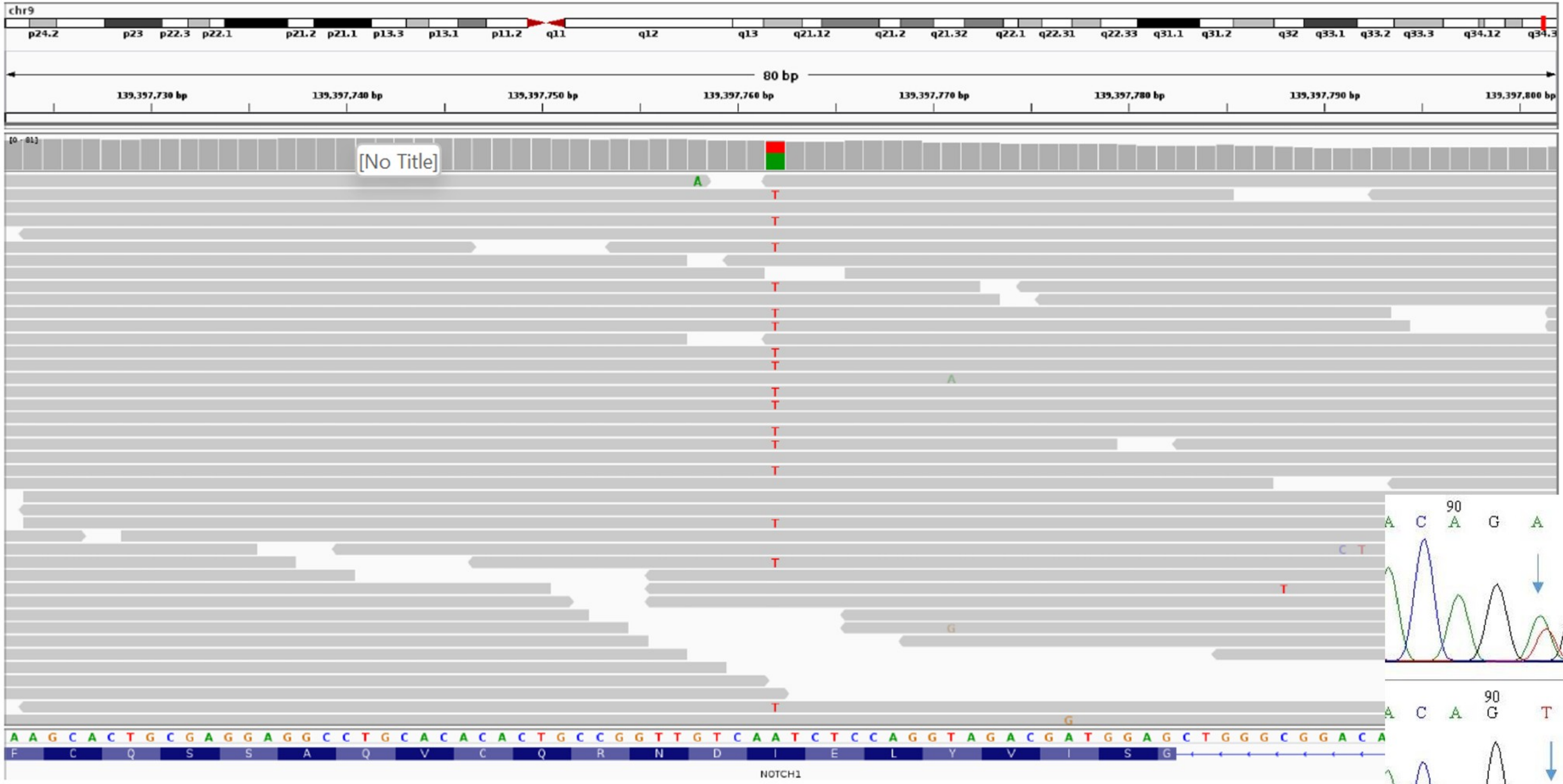




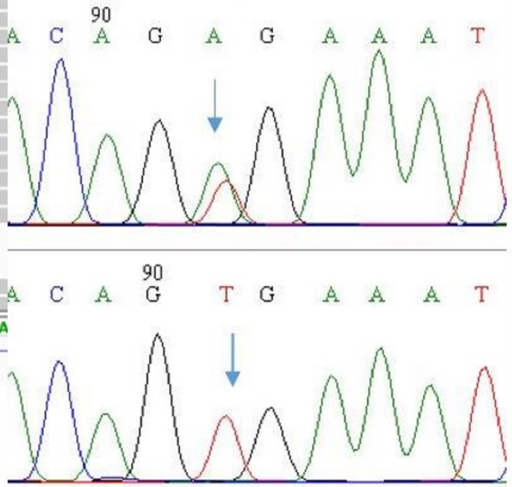


10.5772/intechopen.71769

Trực quan hóa dữ liệu giải trình tự DNA bằng NGS



Sanger



Dữ liệu hệ gen

```
48541 agcccttcaa agaaatgttc tcagcaggca tggagcccag gacttgctcc ctttgggtag
48601 agagccgggt tgaaggtgac tgaagtgaaa tgggacagta gaggcggggg gggtggtgag
48661 ttcctggagg tggggggtgt gggaacctgc tgtgtactga gatgcacccc tgccagttct
48721 gcctgaagat ttgaggcggg gggcaggggg gcggagtgaa gtcattttac tggtaagtaa
48781 ttttaaacct tttaatatta aagcaaacgt ggatatgtaa tgaatgaaat tcattctgga
48841 atgaaaaatt cacgtgatgt tgaaaaataa cacggggctt cagagaggac tttctggctg
48901 gcagcagact ccagattccc agggcccctg caccctcctc tgcccacagg gcaccttaat
48961 ggagaaggtg tgggaggaga gccaggccgg agtcagagca cactggtgac tccacatttg
49021 cagcgtgccc tgcctctctc ctgaggcttg gcaacgtgca atatgctaag caaactcccc
49081 ctgtccccgt ccagtttctg aggacaagag ccaccacctg tagcaaataa agacccagca
49141 accctttgac tcatctttgt gagtctctgg aatcagaggg tagccacatc gctgagaggt
49201 ggagtgaagc actcgggtga aaaggtacaa ggaagtcagg gacaggagtg tggggacatc
49261 acctagacaa tgacagagaa gaggggcaca gccgagtgag gggagagggg ccggcagtcc
49321 tacatcccct ggcctgaagc acgctccagg gcagaaggaa aaacactgtc tttggggtcc
49381 aagagacctg agttcaaatt ctggctccac cactgaccac ctgtgtaacc ttgaactgct
49441 gctgcctgaa cctcaggttc cccttctaaa aatagaggag aaaaggatgc atttctcctt
49501 gcccctgtga gaacgaaatg gtgcaagcac caaggagcct cagcaaaggt cggcctgcc
49561 cccgcctggc caaacctttc ctcttcagga ggccacggca accgtagttt gacagaagag
49621 cagcaccttg atttaatgct tcccagcatg tgtccttgag caagtcacct aacctctctg
49681 ggctgcttcc tcattgggaa aatatggctg ccagtaaaac ctgccctgtc cacctcctgg
49741 ggcacttggc aaacagcaaa agagtccaaa tgtgcaggct gggccaggcg cagtggctca
49801 tgcctgtaat cccagcaatt taggaagcca aggtgggcgg atcacctgag gtcaggagtt
49861 tgagaccagc ctggccaaca tggtgaaacc ttgtctctac aaaaatacaa aaattagccg
49921 ggcatgatgg cgggtgcctg taatcccagt tactcgggag gctgaggcaa gagaatcgct
49981 tgaacccgga aggggaaggt tgcagtgagc caagattgtg ccactgcact ccagcctggg
50041 caacagagcg agactctgtc tcaaaaaaaa aaaaaaaaaa aaacaatgca gagctggctg
50101 tgtaaaaaac ctgttccact gcagggccca gtgtccacca ggctggggtg caggcctatg
50161 gggtgggggc ccagcatcag cctctcagca gccctgggag gcggggcgca tcccgtgccc
50221 ctcgtggtct ggatgtgttc tagcccaagt cctaggttac acctgccgtc gcctggcctc
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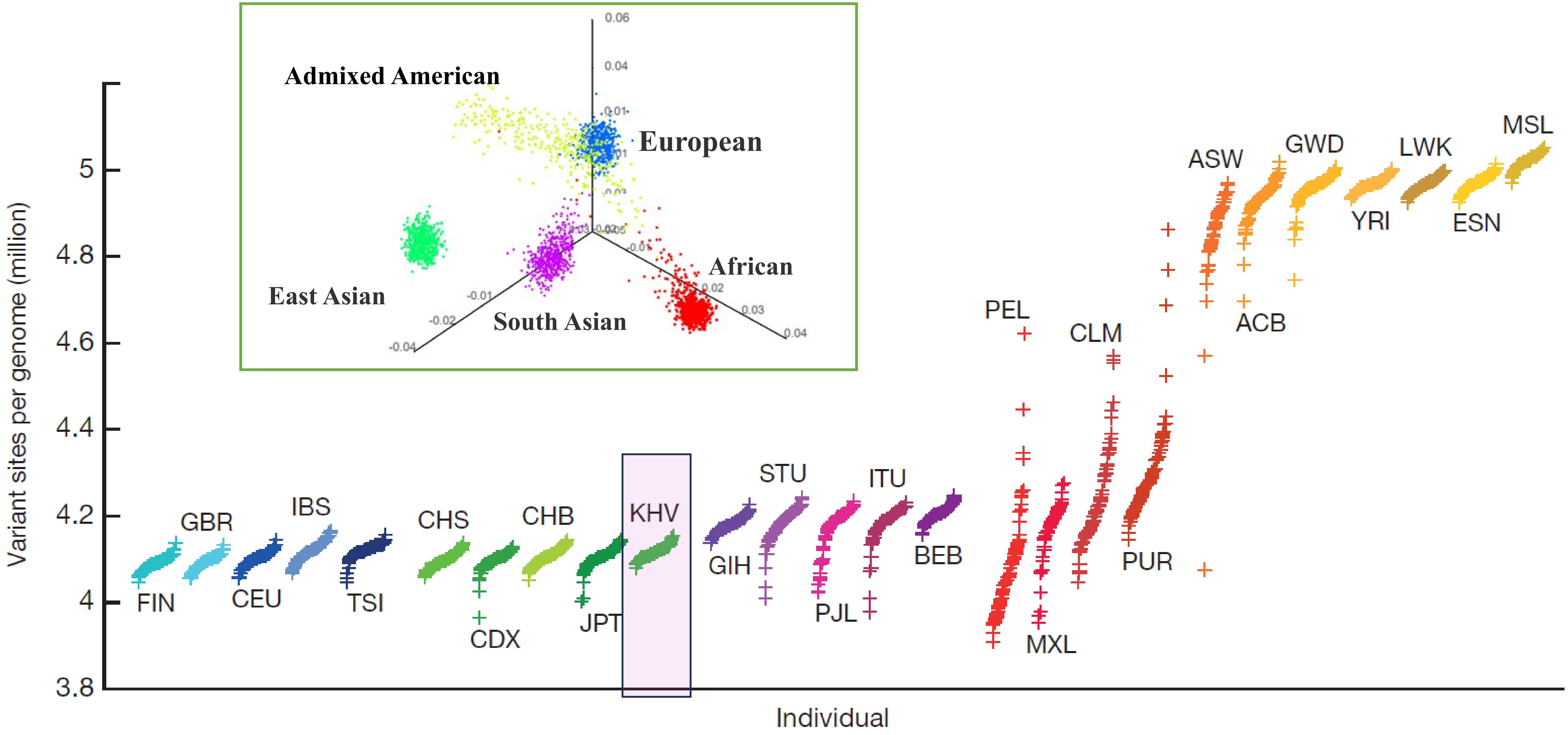
50461 agcccttgtg gtcgcaggtg ctggtcctgg cctcttcccc tcccatgcag gagcacagga
50521 gagatggctt ctgaggacct gttgcagctg tggccctggg aatagatttg ccagggagct
50581 ttaaagcagc tgagtgtgtc atccagctaa gcctggggaa ggagcttggc tcaggtcctg
50641 acaggtgtga cagggatggg gactgggaag taagagatga aaccctggct ggaggctgtg
50701 agcttccaca gccagcgctg gacagggagg gtccagatat acccactagt gccctcacca
```

https://www.youtube.com/watch?v=mDi-crPgjFw&t=3s

Human Genome Variation: 1000 Genomes Project

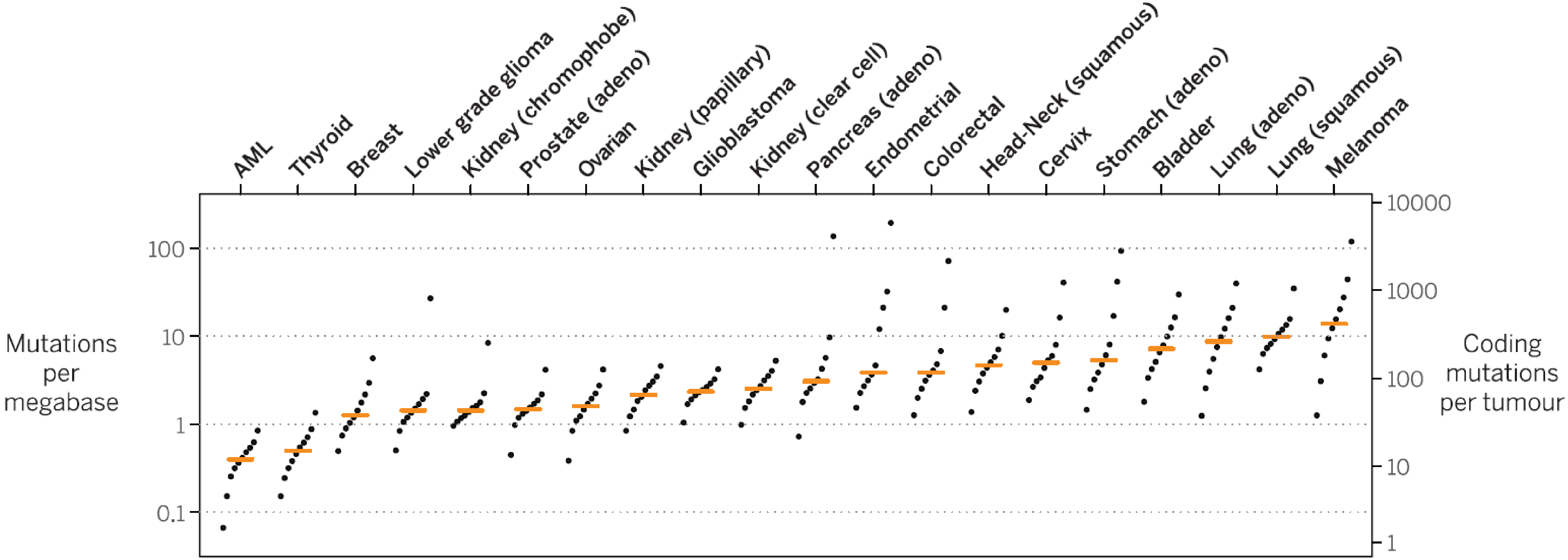


The number of variant sites per genome of 1K human genomes project (2015)
Kinh in Ho Chi Minh City, Vietnam (KHV)

doi:10.1038/nature15393

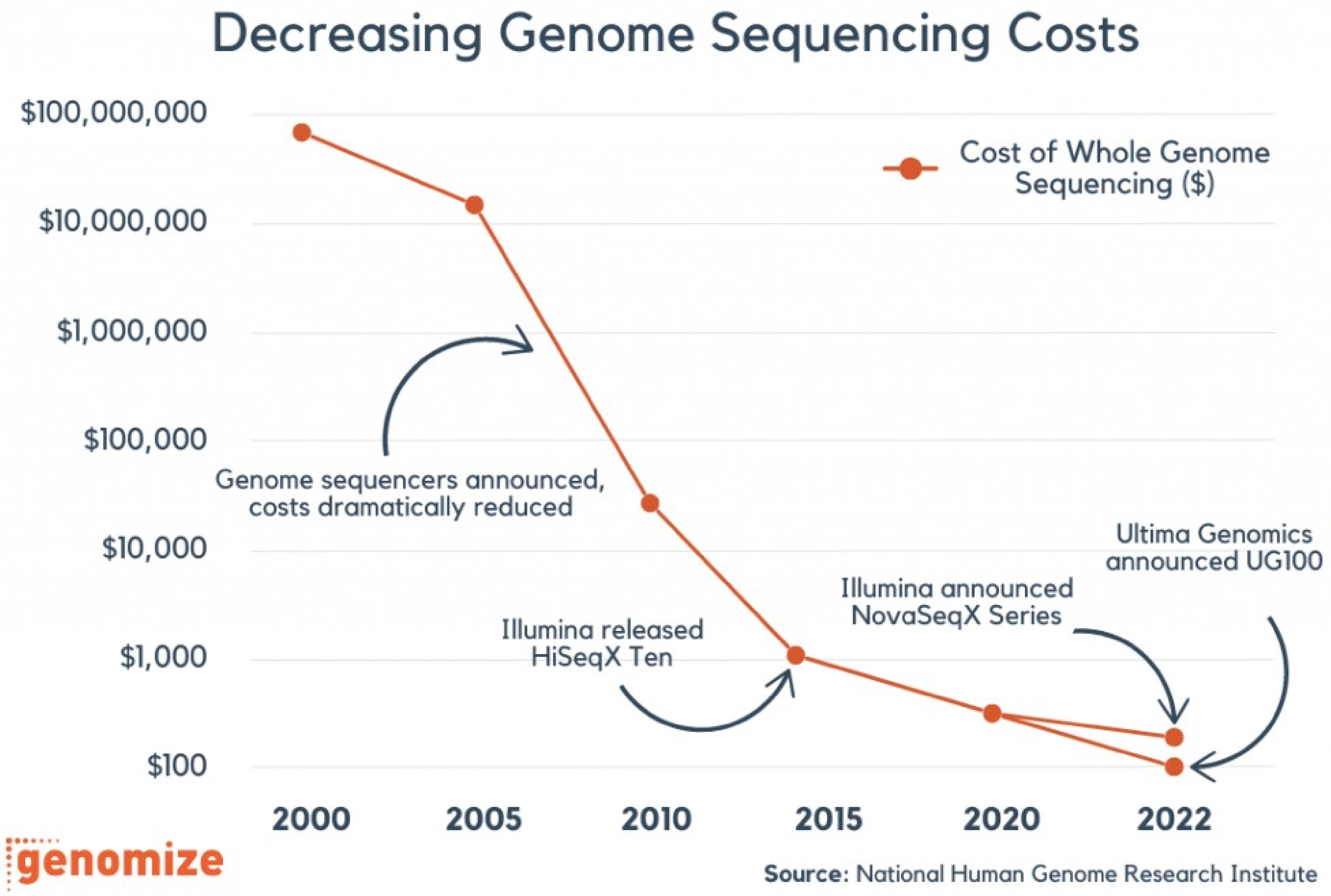
https://bwlewis.github.io/1000_genomes_examples/PCA_overview.html

Cancer Genome Somatic Variation

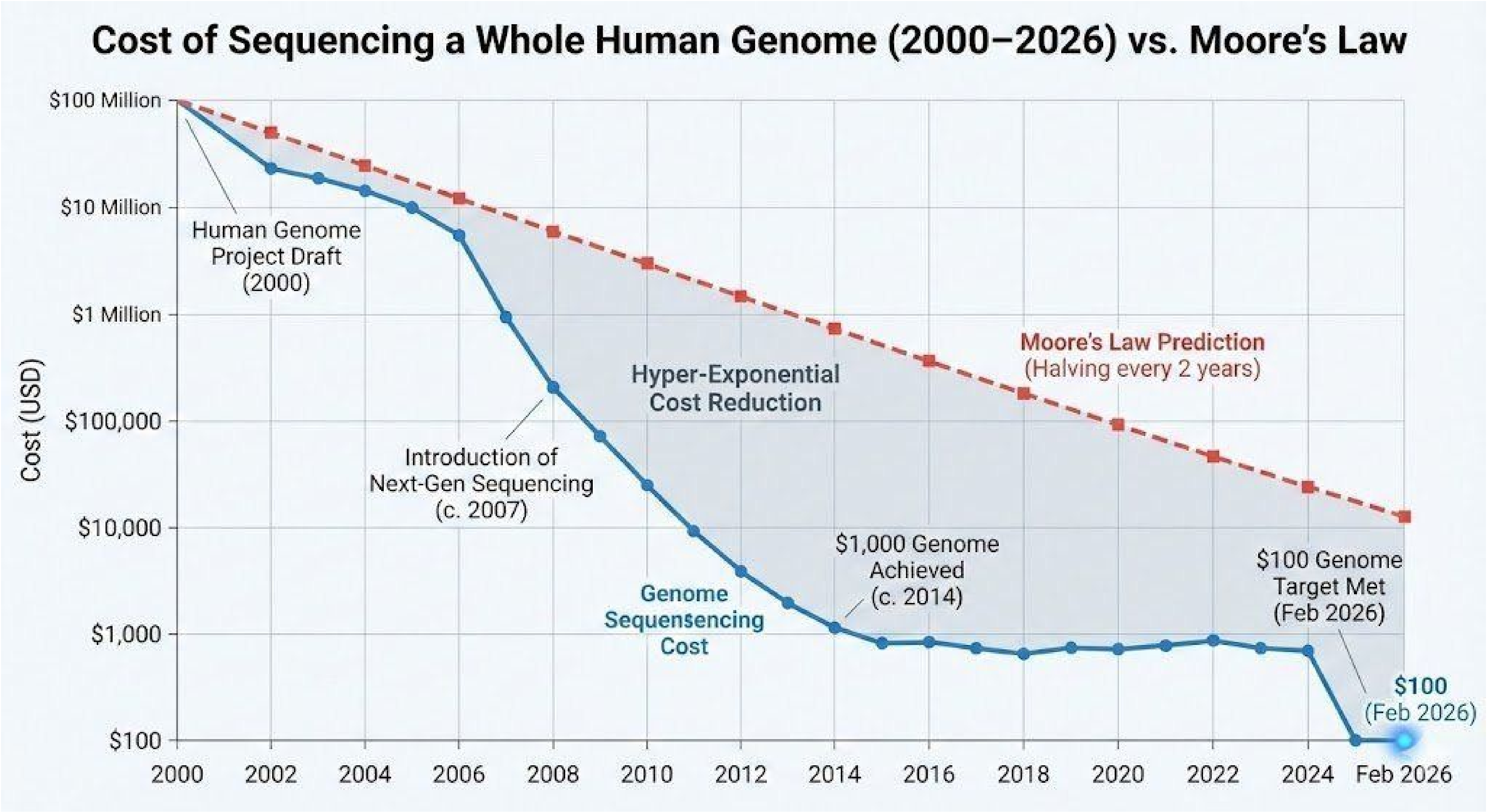


Mutation burden in 20 tumor types and relative contribution of different mutational processes.
For each tumor type, samples were divided into deciles on the basis of their mutation burden. (2015)

DOI: 10.1126/science.aab4082



Decreasing Genome Sequencing Costs
$100,000,000
$10,000,000
$1,000,000
$100,000
$10,000
$1,000
$100
2000
2005
2010
2015
2020
2022
Cost of Whole Genome Sequencing ($)
Genome sequencers announced, costs dramatically reduced
Illumina released HiSeqX Ten
Illumina announced NovaSeqX Series
Ultima Genomics announced UG100
genomize
Source: National Human Genome Research Institute

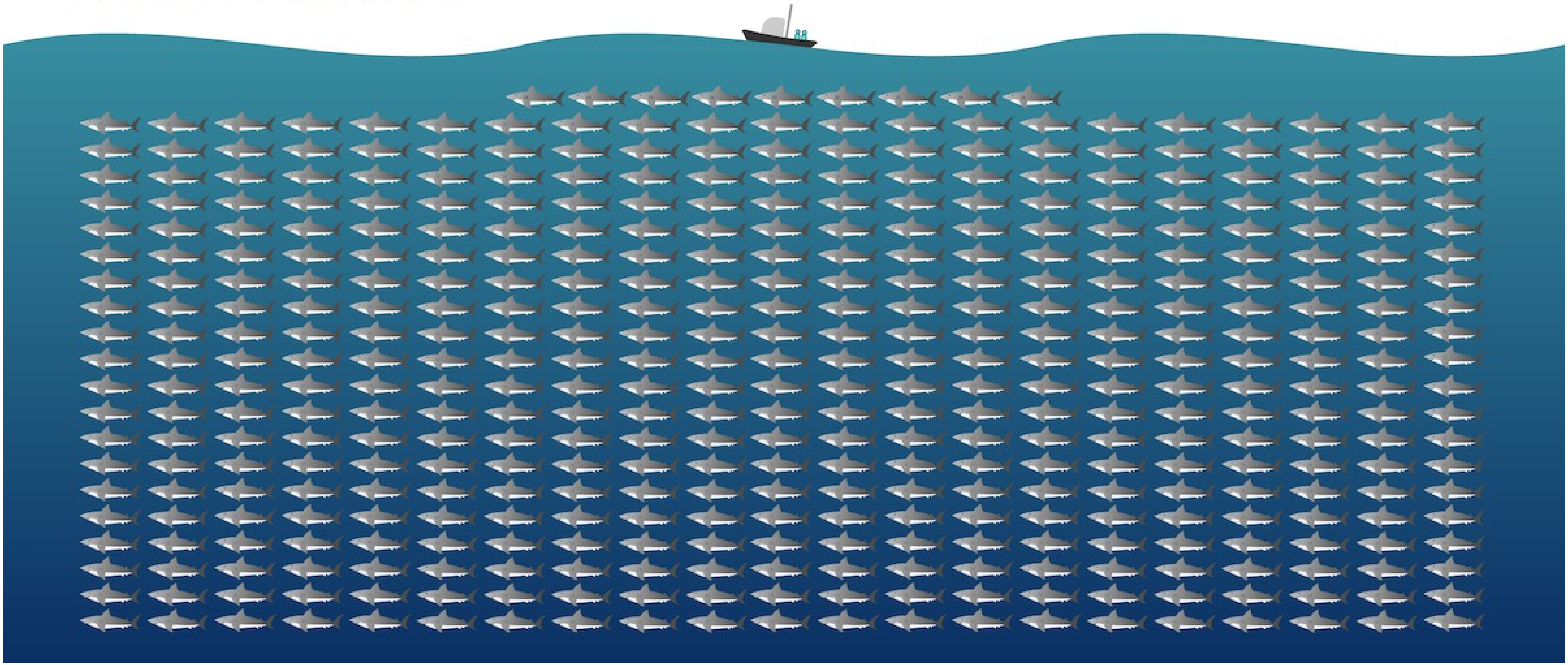

Cost of Sequencing a Whole Human Genome (2000–2026) vs. Moore's Law
$100 Million
$10 Million
$1 Million
$100,000
$10,000
$1,000
$100
Cost (USD)
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Feb 2026
Human Genome Project Draft (2000)
Introduction of Next-Gen Sequencing (c. 2007)
Hyper-Exponential Cost Reduction
Moore's Law Prediction (Halving every 2 years)
Genome Sequensencing Cost
$1,000 Genome Achieved (c. 2014)
$100 Genome Target Met (Feb 2026)
$100 (Feb 2026)

	Giải trình tự theo kiểu cũ	Giải trình tự thế hệ mời (NGS)
Platform	Sanger 	**SURFSeq Q** 
Số bộ gen người	**12**	**20,000 (54)**
Thời gian	~10 năm (1990 đến 2000)	1 năm (1 ngày)
Giá thành	~3 tỉ USD	~200 USD
Chất lượng	Không xác định	Q30 > 85%
Chứng nhận	Không xác định	IVD

How big is 40 exabytes?

Genomics projects will generate 40 exabytes of data in the next decade.

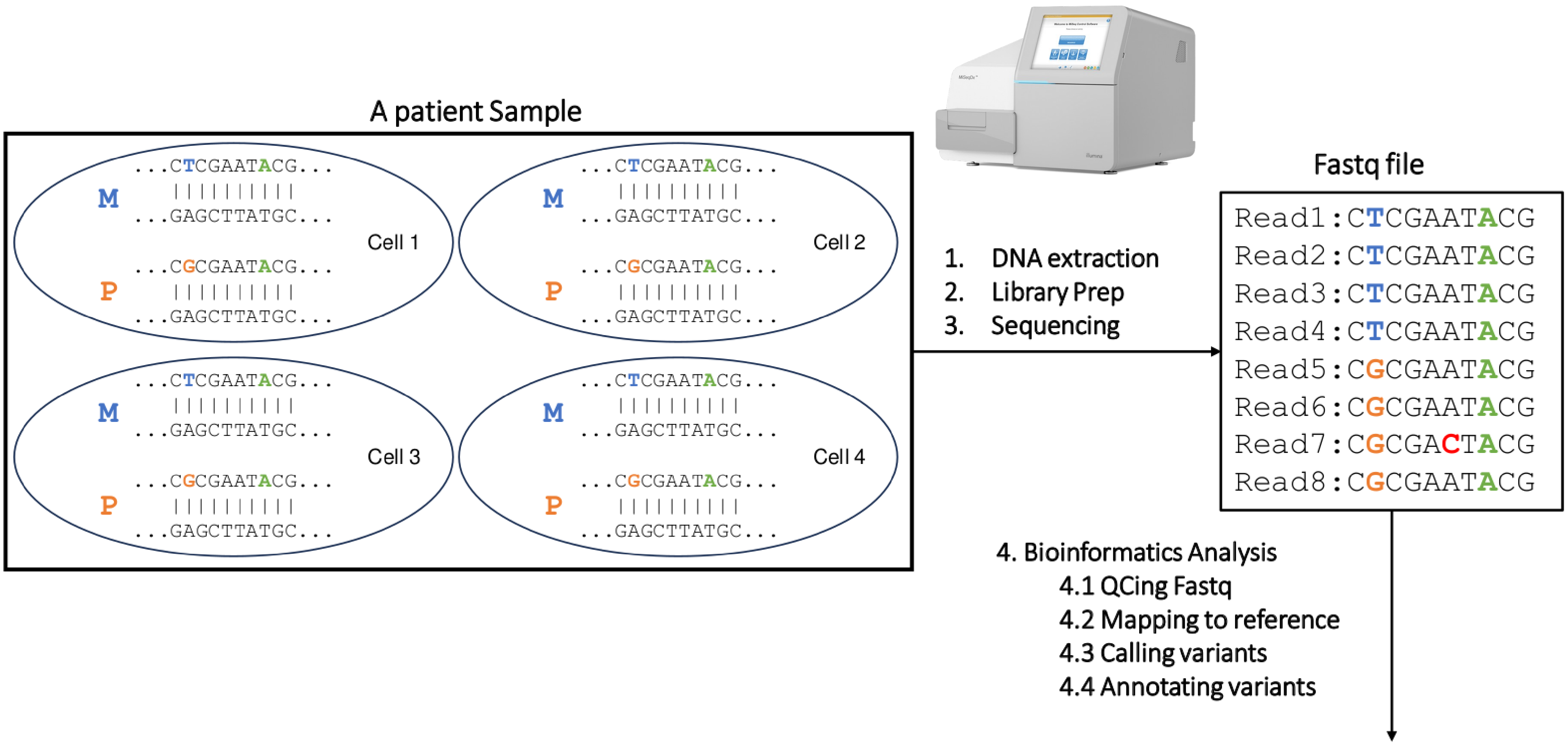
Each shark = 100,000,000 GB of data



https://www.genome.gov/about-genomics/fact-sheets/Genomic-Data-Science

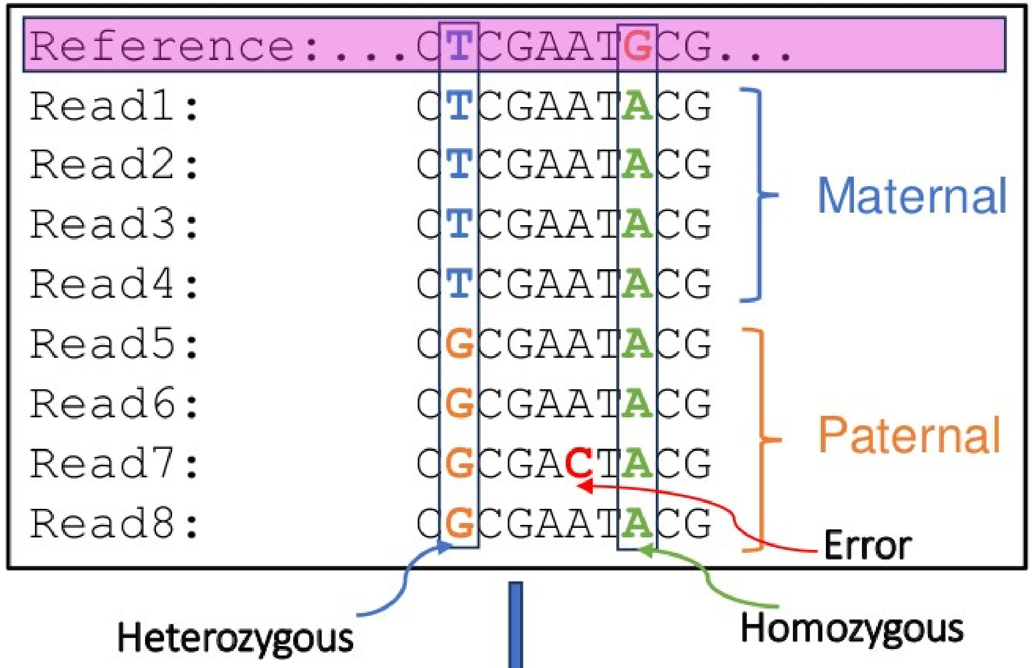
Quy trình XÉT NGHIỆM gen bằng phương pháp giải trình tự thế hệ mới (NGS)

Các bước trong XÉT NGHIỆM gen bằng phương pháp giải trình tự thế hệ mới (1)



Các bước trong XÉT NGHIỆM gen bằng phương pháp giải trình tự thế hệ mới (2)

4.2 Mapping reads to reference



ANN=G|stop_gained|HIGH|OR4F5|ENSG00000186092|transcript|ENST0 0000641515.2|protein_coding|3/3|c.822T>G|p.Trp274*|882/2618|822/ 981|274/326||Pathogenic

ANN=A|frameshift_variant|HIGH|ZSWIM2|ENSG00000163012|transcrip t|ENST00000295131.3|protein_coding|9/9|c.1238G>A|p.Ile413|1293/ 2451|1238/1902|413/633||;LOF=(ZSWIM2|ENSG00000163012|1|1.00)

4.4 Annotating variants

4.3 Calling variants

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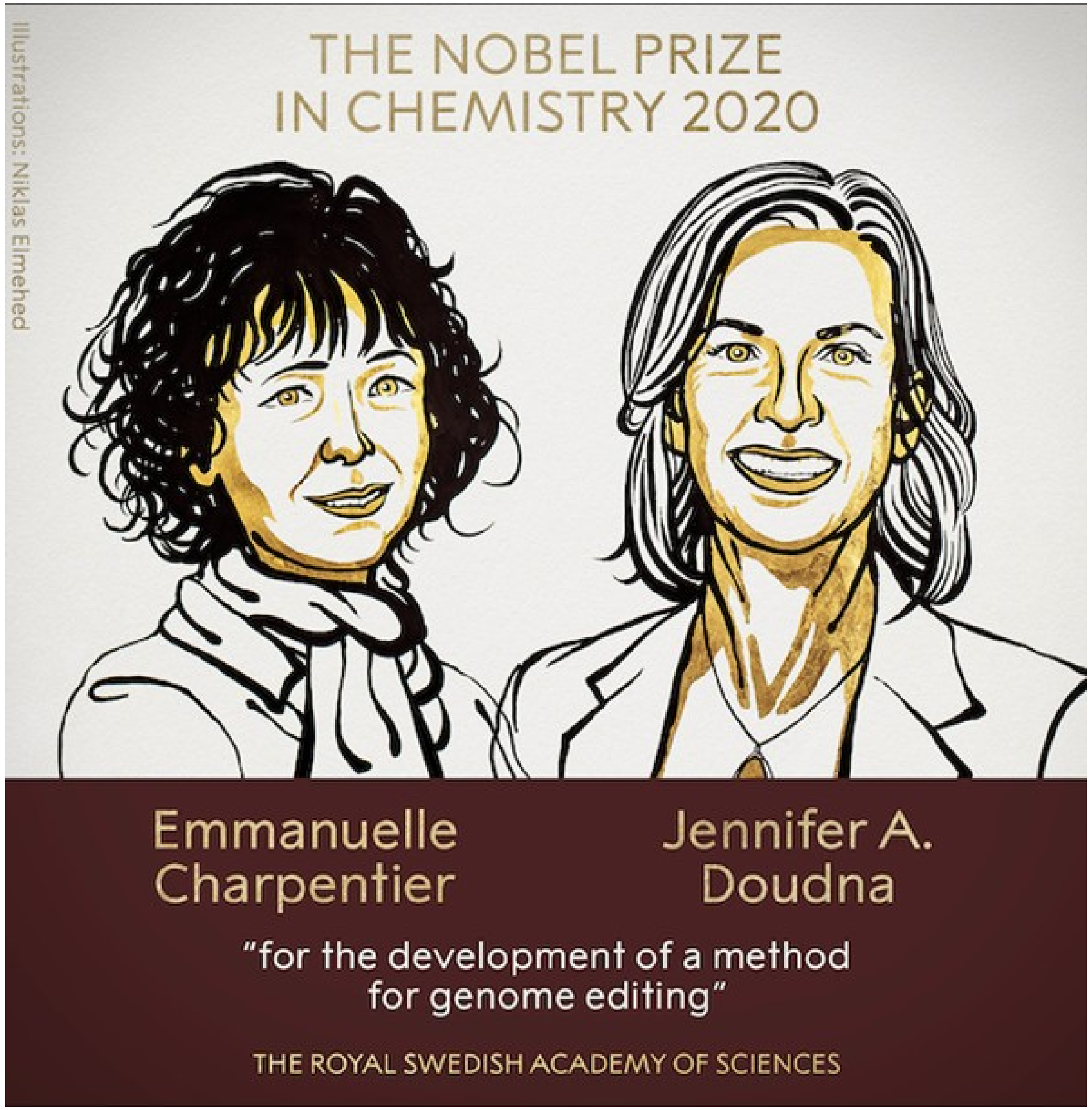
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=AD,Number=2,Type=Integer,Description="Read depth for each allele">
#CHROM  POS    ID         REF  ALT  QUAL  FILTER  FORMAT       Sample1
20      14370  rs6054257  T    G    129   PASS    GT:GQ:DP:AD  0/1:48:8:4,4
20      17330  .          G    A    150   PASS    GT:GQ:DP:AD  1/1:49:8:8,8
```

Hội đồng Hệ Gen: phiên giải và hội chẩn những biến thể trong báo cáo kết quả NGS

https://ggba.swiss/en/the-first-medical-genomics-center-opens-in-geneva/

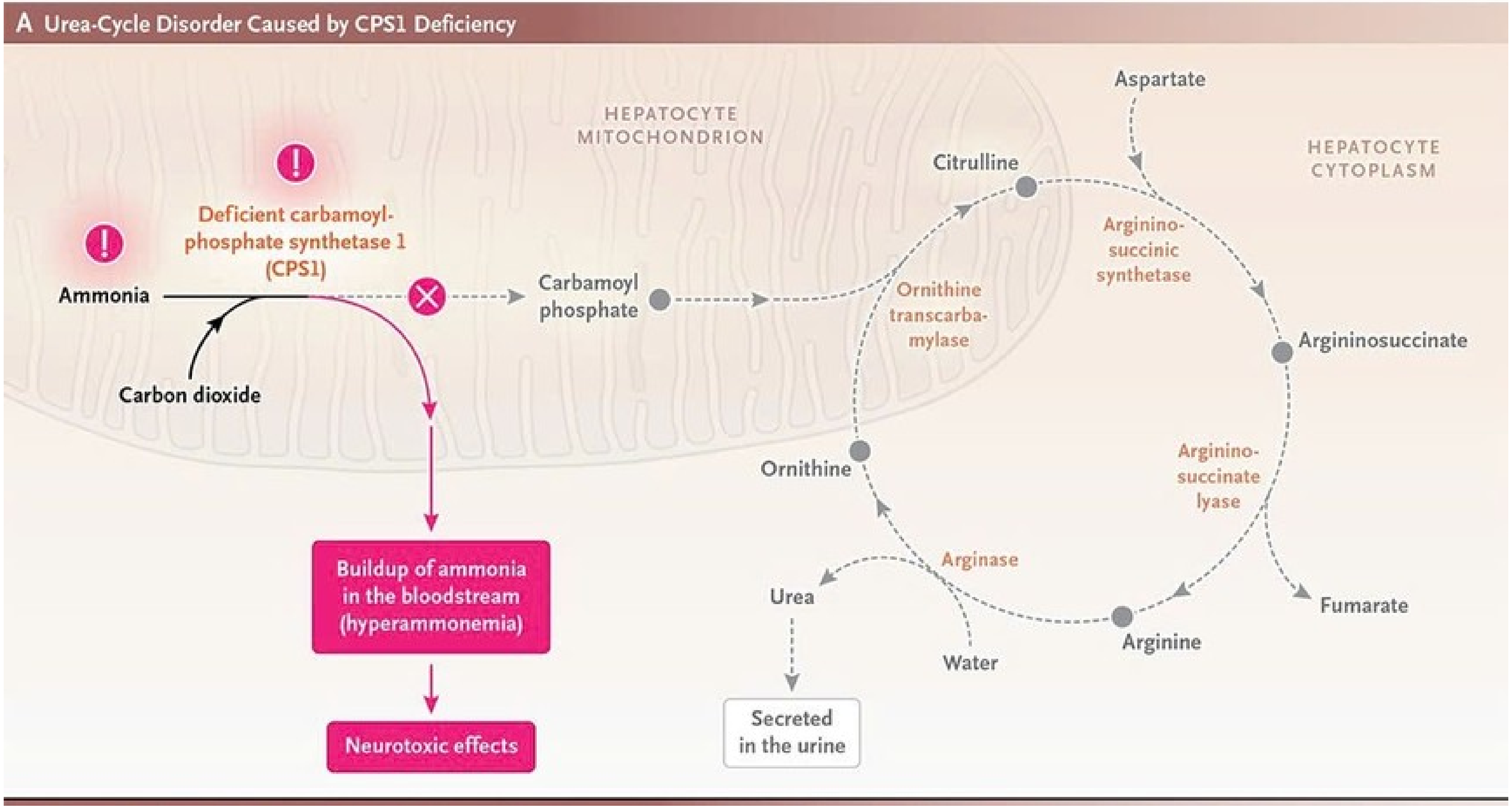
LIỆU CÓ GIÚP ÍCH?



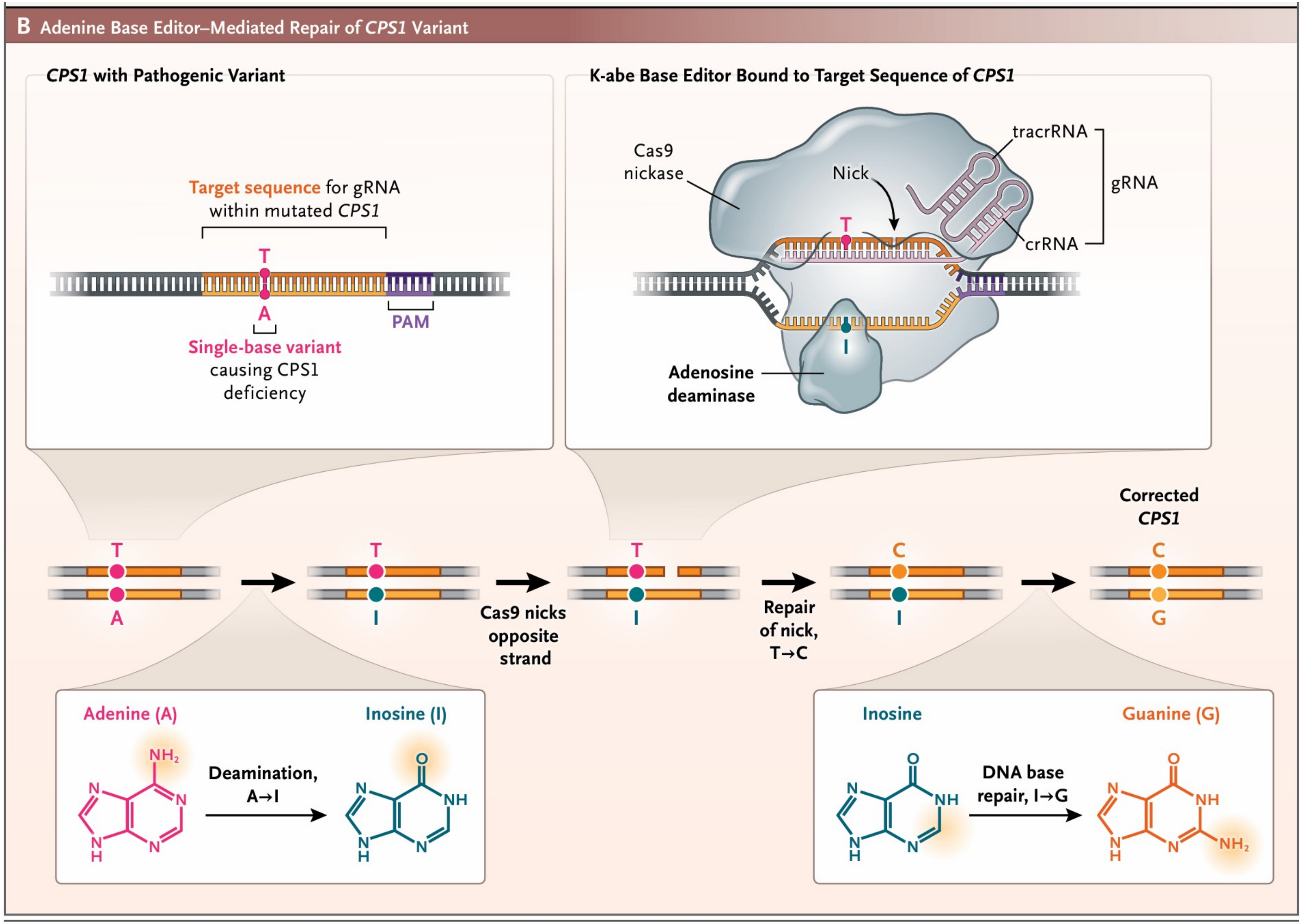
CRISPR-CAS9

Illustration by Niklas Elmehed. Source: nobelprize.org.

New England Journal of Medicine (NEJM), May 2025



Gropman AL, Komor AC. doi: 10.1056/NEJMe2505721. Epub ahead of print. PMID: 40373054.



Gropman AL, Komor AC. doi: 10.1056/NEJMe2505721. Epub ahead of print. PMID: 40373054.

NEWS | HEALTH

Gene-editing therapy made in just 6 months helps baby with life-threatening disease

Custom CRISPR paves the way for treating genetic disorders in tailormade ways

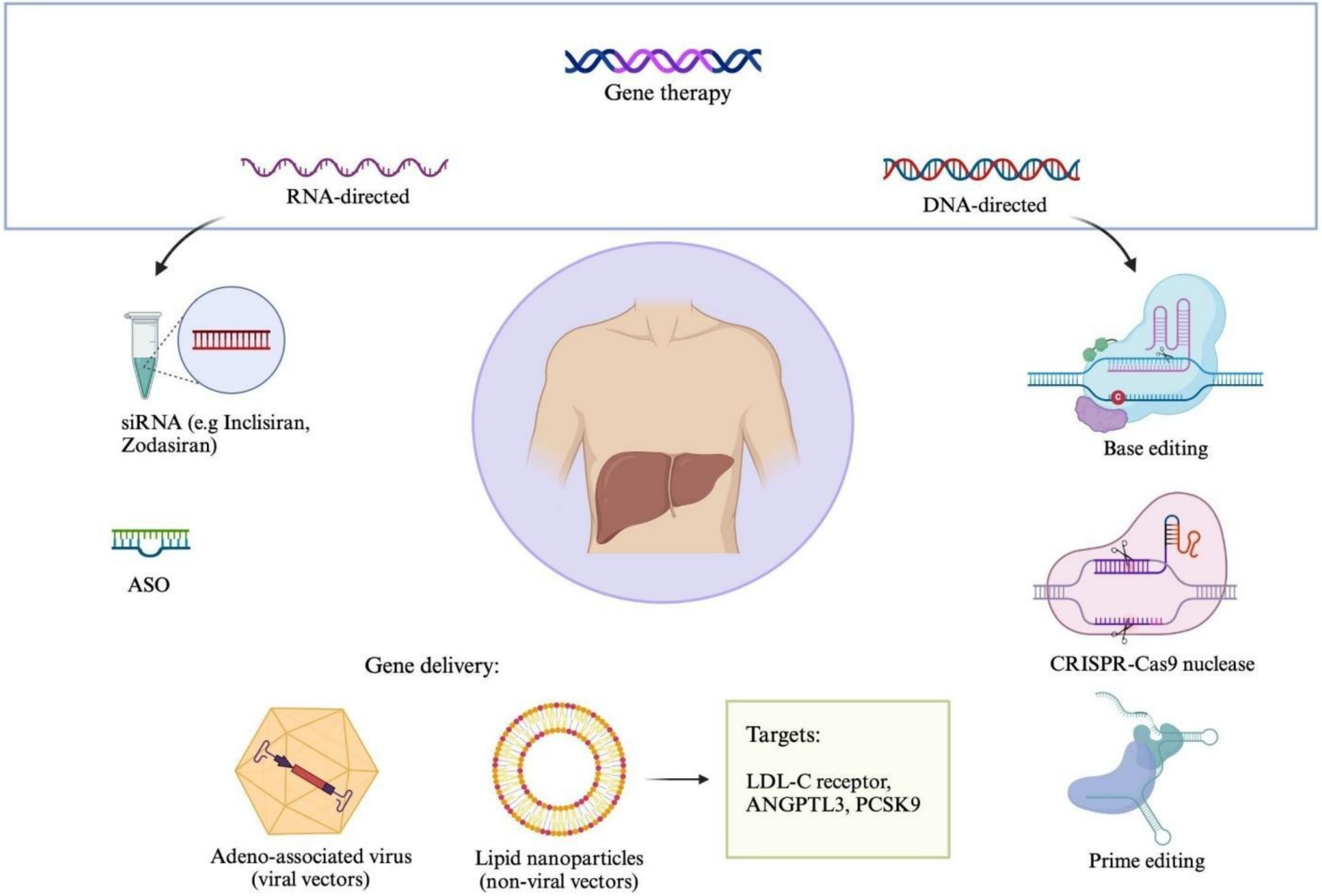
15 MAY 2025 · 1:00 PM ET · BY JOCELYN KAISER

May 2025

An infant who received a customized gene-editing treatment, shown here with researchers Kiran Musunuru (left) and Rebecca Ahrens-Nicklas, now needs less medicine to defuse a blood buildup of ammonia. . From "Gene-editing therapy made in just 6 months helps baby with life-threatening disease" by Jocelyn Kaiser, 2023. https://www.science.org/content/article/gene-editing-therapy-made-just-6-months-helps-baby-life-threatening-disease

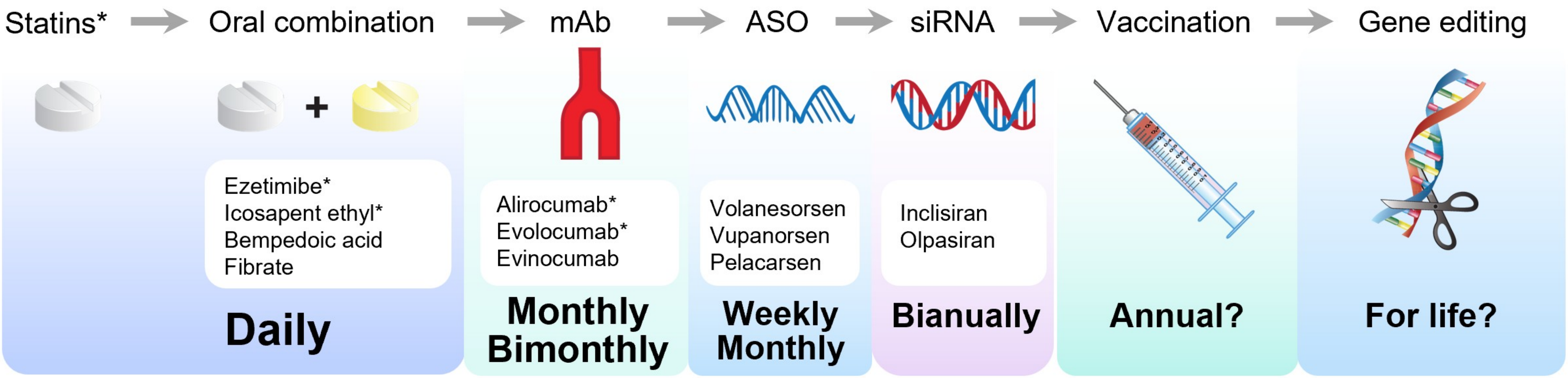
Tăng cholesterol máu gia đình (Familial Hypercholesterolemia - FH)

Bảng gen cho bệnh tăng cholesterol máu gia đình: APOB, LDLR, LDLRAP1 và PCSK9



Madelyn Hurwitz, Expert Analysis, May 27, 2025

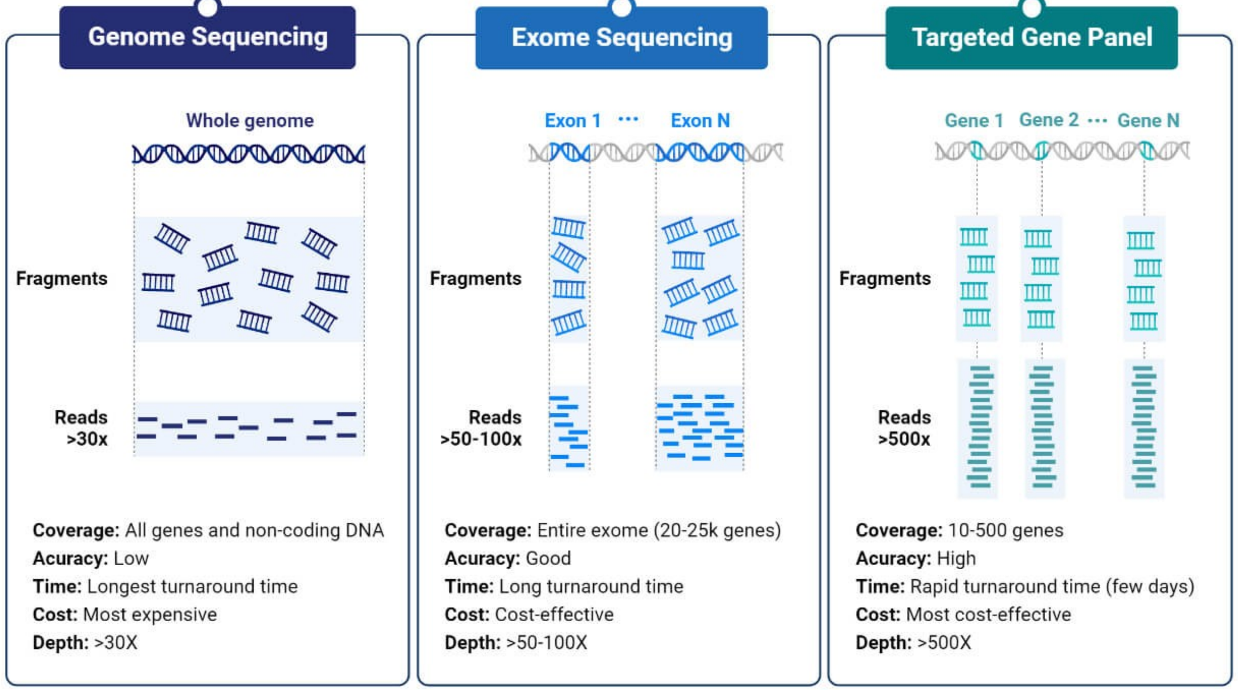
Điều trị tăng cholesterol máu gia đình
(Familial Hypercholesterolemia - FH)



Từ slide GS TS. Trương Quang Bình - ĐHYD TP HCM

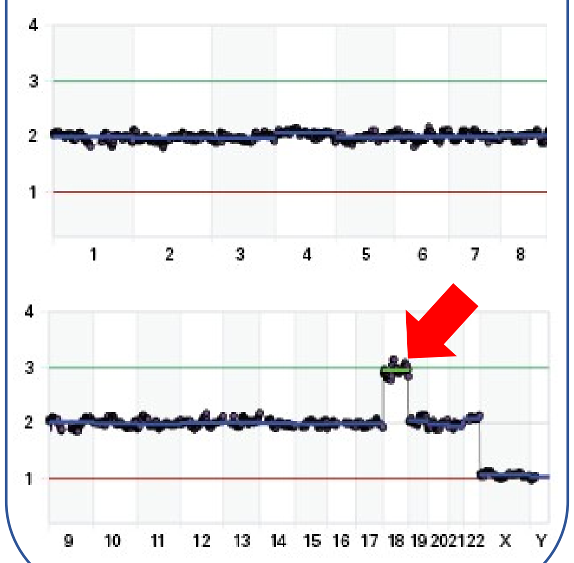
Ứng dụng giải trình tự gen thế hệ mới trong lâm sàng

Single Nucleotide Variant (SNV)

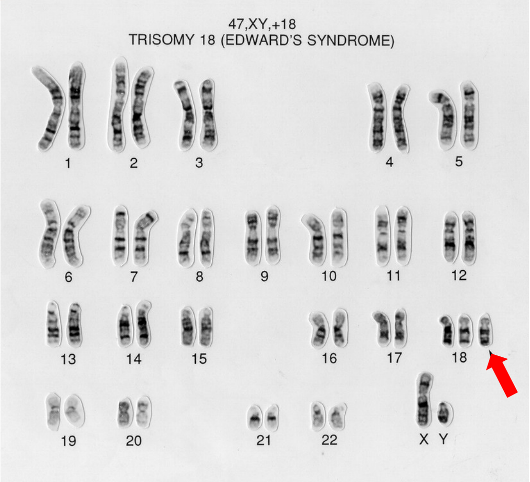


https://microbenotes.com/next-generation-sequencing-ngs/

Copy Number Variation (CNV)
Low-Pass Genome Sequencing



Karyotype



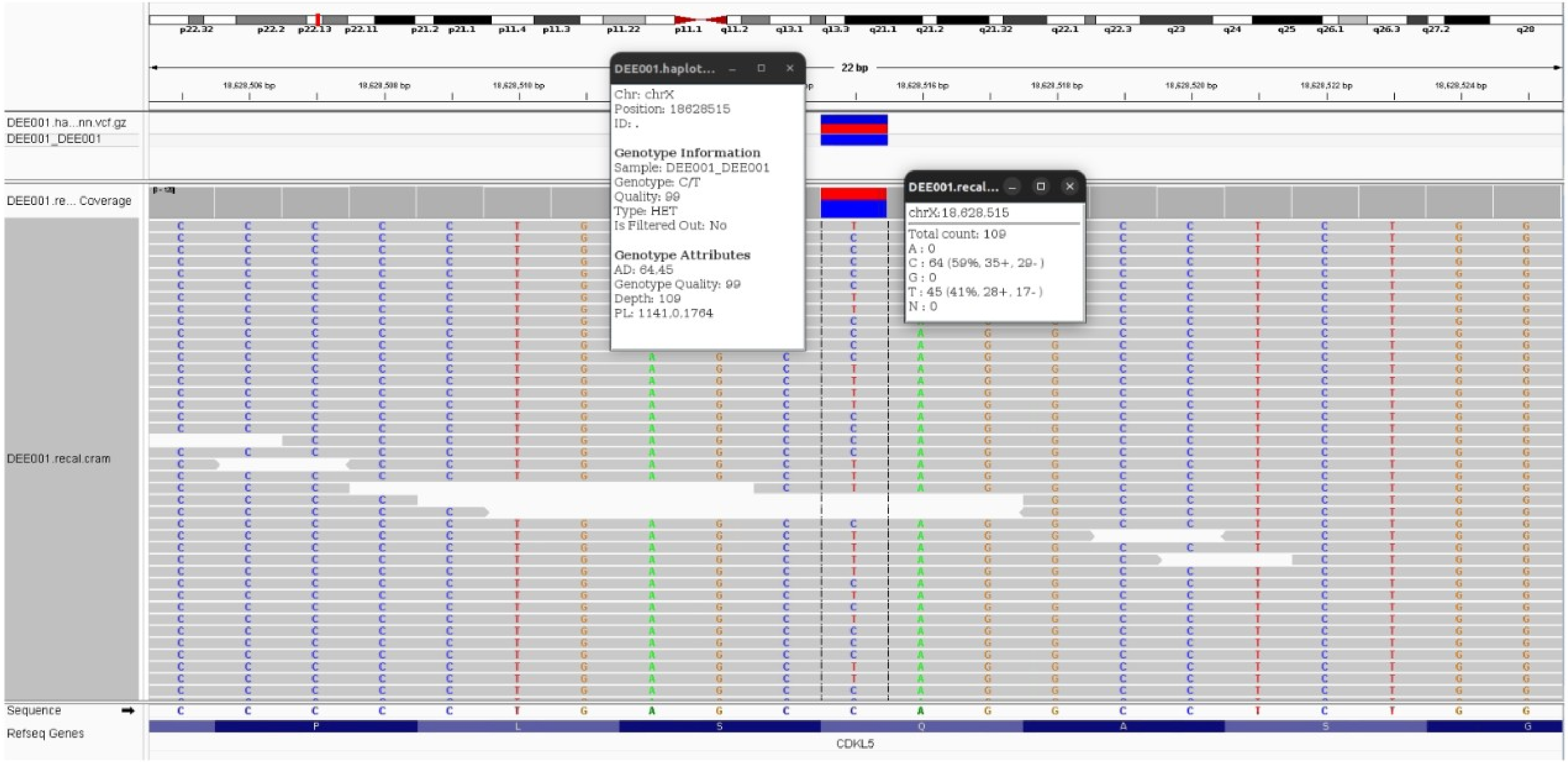
https://wellcomecollection.org/search/images?query=eaahzt2u#

Năm ví dụ về XÉT NGHIỆM gen cho BỆNH DI TRUYỀN và UNG THƯ

Ví dụ 1: Phát hiện sớm bệnh động kinh bằng xét nghiệm gen (1)

Whole Exome Sequencing - WES

IGV - DEE001 – **CDKL5**

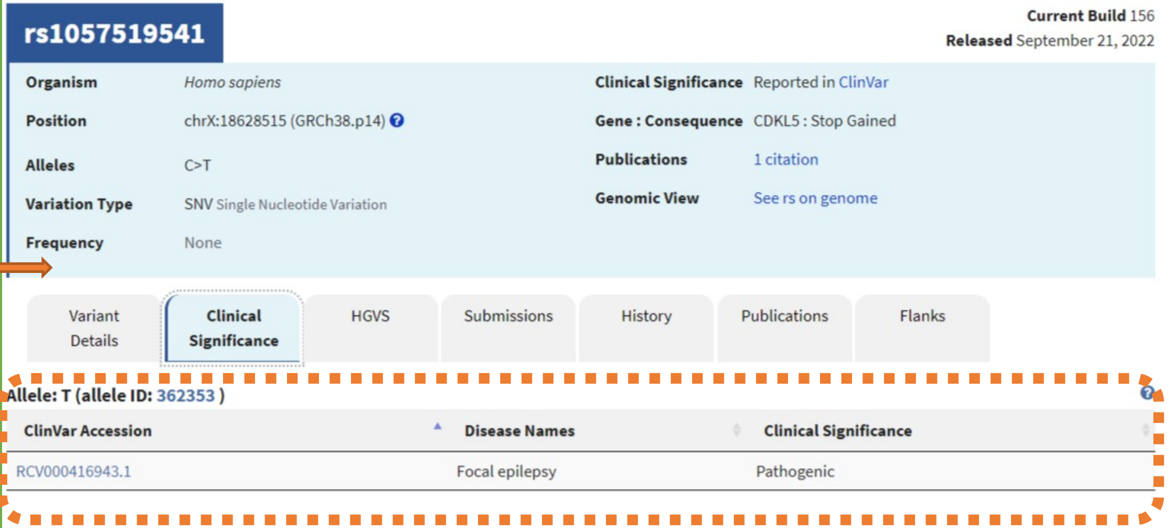


Ví dụ 1: Phát hiện sớm bệnh động kinh bằng xét nghiệm gen (2)
Whole Exome Sequencing - WES

DEE001

Gene	Amino acid change	cDNA	Variant type	Allele frequency	Transcript	Variant effect	ClinVar significance
CPT2	p.Arg631Cys	c.1891C>T	SNP	0.5	ENST00000371486.4	Missense variant	Pathogenic
CDKL5	p.Gln881Ter	c.2641C>T	SNP	0.5	ENST00000623535.2	Stop gained (Nonsense)	Pathogenic
GALC		c.1162-4del	DEL (1bp)	1	ENST00000261304.7	Intron variant	Conflicting interpretations of pathogenicity
TUBB2B	p.Ala248Val	c.743C>T	SNP	0.5	ENST00000259818.8	Missense variant	Conflicting interpretations of pathogenicity

DEE001 – **CDKL5** (Xp22.13)



https://www.ncbi.nlm.nih.gov/snp/rs1057519541#clinical_significance

NGUYEN Thuy-Minh-Thu, MD
Nguyen Le Duc Minh, MD

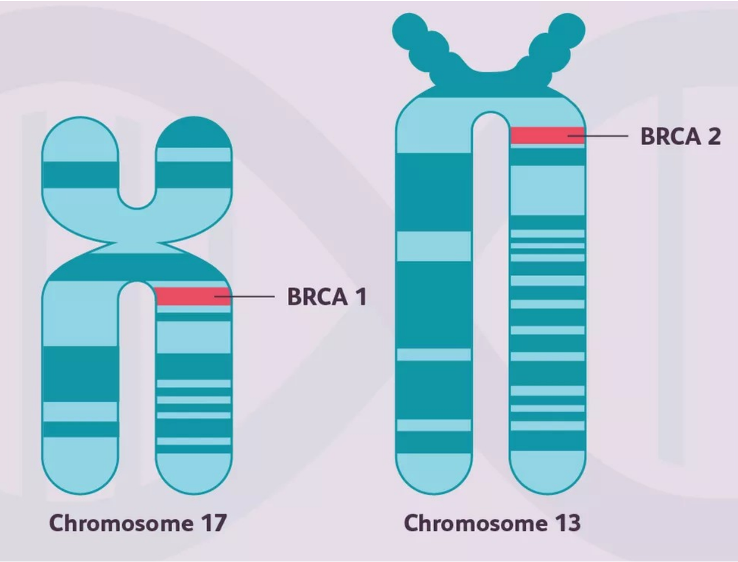
Ví dụ 2: Sàng lọc bệnh ung thư bằng xét nghiệm gen Gene Panel – 53 genes của MGI

Gene	Amino Acid Change	Coding	Variant Type	Allele Frequency	Transcript	Variant effect	ClinVar Significance
CTNNB1	p.Ser33Tyr	c.98C>A	SNP	0.5	ENST00000349496	MISSENSE	Pathogenic/ Likely_pathogenic
PIK3CA	p.Gly914Arg	c.2740G>A	SNP	0.5	ENST00000263967	MISSENSE	Pathogenic
KRAS	p.Gly12Asp	c.35G>A	SNP	0.5	ENST00000256078	MISSENSE	Pathogenic
BRCA2	p.Ile2675AspfsTer6	c.8021dup	INS	0.5	ENST00000544455	FRAMESHIFT	Pathogenic

Nguyen Le Duc Minh, MD

Ví dụ 3: Hỗ trợ điều trị bệnh ung thư vú bằng xét nghiệm gen Gene Panel BRCA1 và BRCA2

Olaparib (AstraZeneca) là một loại thuốc dùng để duy trì điều trị ung thư vú, buồng trứng, tuyến tiền liệt và tuyến tụy giai đoạn tiến triển có đột biến BRCA ở người lớn.



KẾT QUẢ XÉT NGHIỆM *BRCA1/2*

Họ và tên : TRẦN THỊ X. Tuổi : 1956 Giới tính : NỮ

Số hồ sơ:

KHOA: BS điều trị:

Bệnh phẩm : **Mô vùi nến** Số block: **XXXX**

Yêu cầu: **Xét nghiệm giải trình tự gen trên hệ thống Miseq [02 gen *BRCA1* và *BRCA2*]**

Ngày nhận chỉ định: **20/10/2022** Ngày thực hiện: **25/10/2022**

Chẩn đoán lâm sàng: Ung thư buồng trứng dịch trong grade cao/ Ung thư vú trái

Chất lượng mẫu: MẪU ĐẠT (kích thước 17mm x 16mm, thành phần bướu 70%)

Phương pháp: Giải trình tự gen bằng phương pháp NGS cho 02 gen *BRCA1* và *BRCA2*

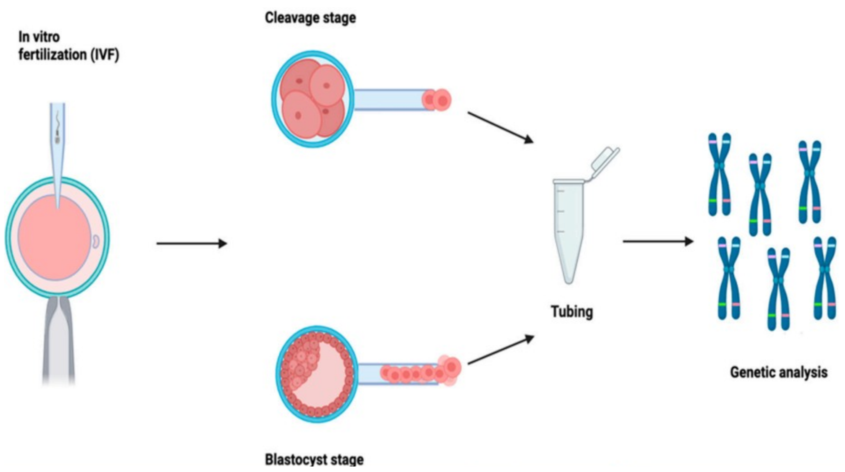
- Hệ thống xét nghiệm **Illumina MiSeqDx (CE/US-IVD)**
- Bộ xét nghiệm: **NGeneBio BRCAaccuTest™Plus (CE-IVD)**
- Phần mềm hỗ trợ phân tích kết quả: **NGeneBio NGeneAnalySys™ (CE-IVD)**

Kết quả: **PHÁT HIỆN 1 BIẾN THỂ MẤT ĐOẠN NUCLEOTIDE NHỎ (DEL) GÂY BỆNH TRÊN GEN *BRCA1***

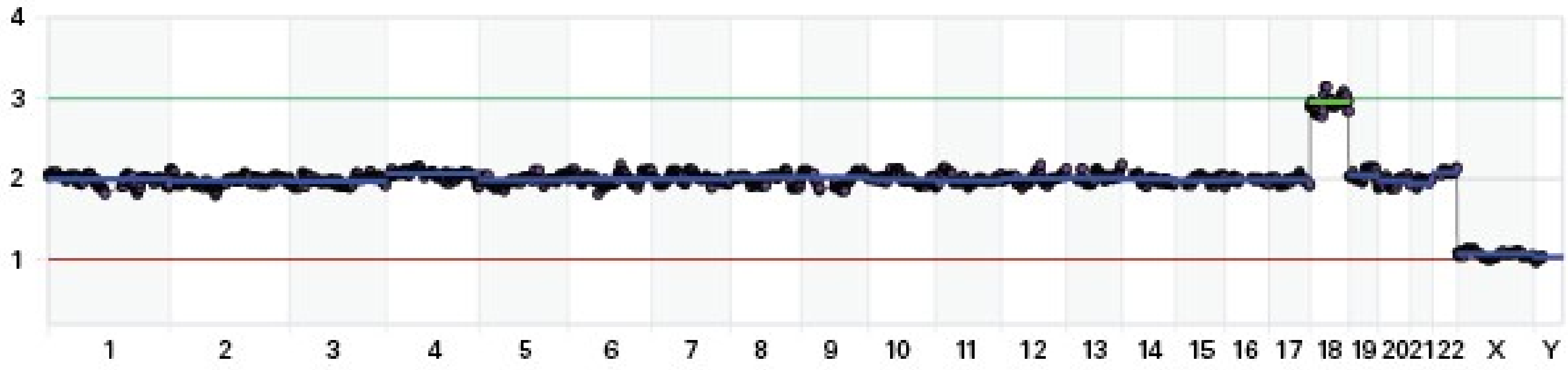
MÔ TẢ KẾT QUẢ			
Gen	**Biến thể gây bệnh/có khả năng gây bệnh**	**Tỷ lệ**	**Phân loại**
BRCA1	**c.5335del (p.Gln1779AsnfsTer14)**	**82.03%**	**Gây bệnh (Pathogenic)**
BRCA2	**Không phát hiện**	**Không**	**Không**

Ví dụ 4: Sàng lọc phôi trong hỗ trợ sinh sản IVF

Xét nghiệm tiền làm tổ PGT-A



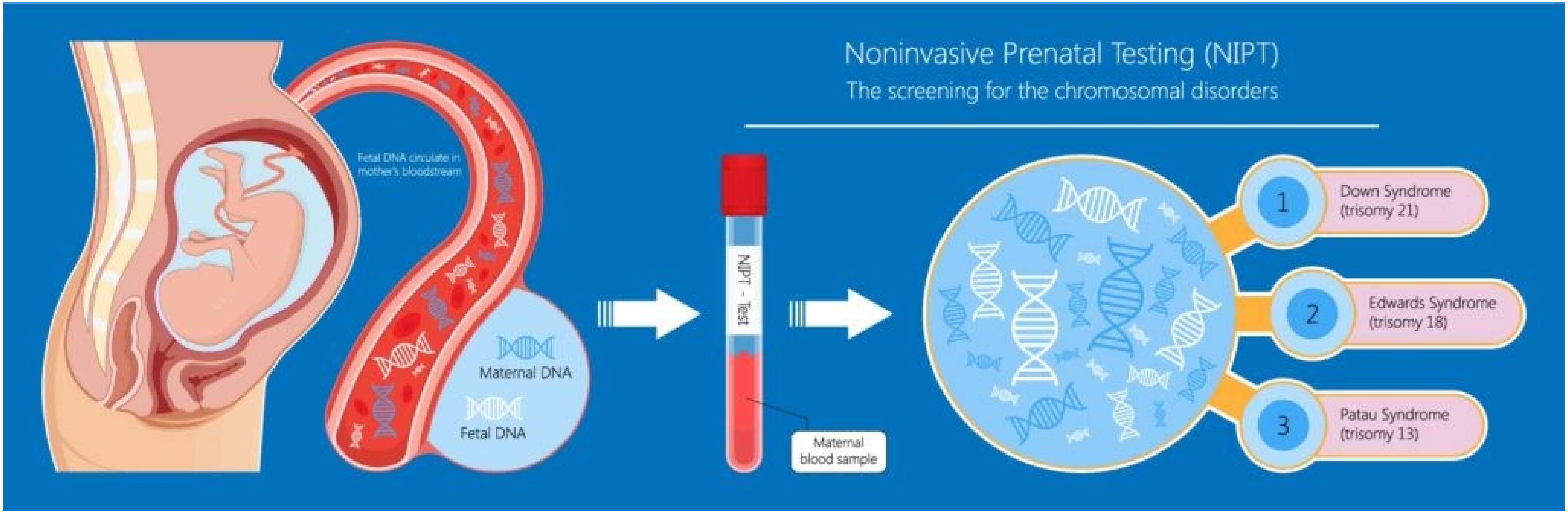
10.3390/genes14112095



Mẫu phôi có 3 nhiễm sắc thể 18 (trisomy) trong DNA hệ gen

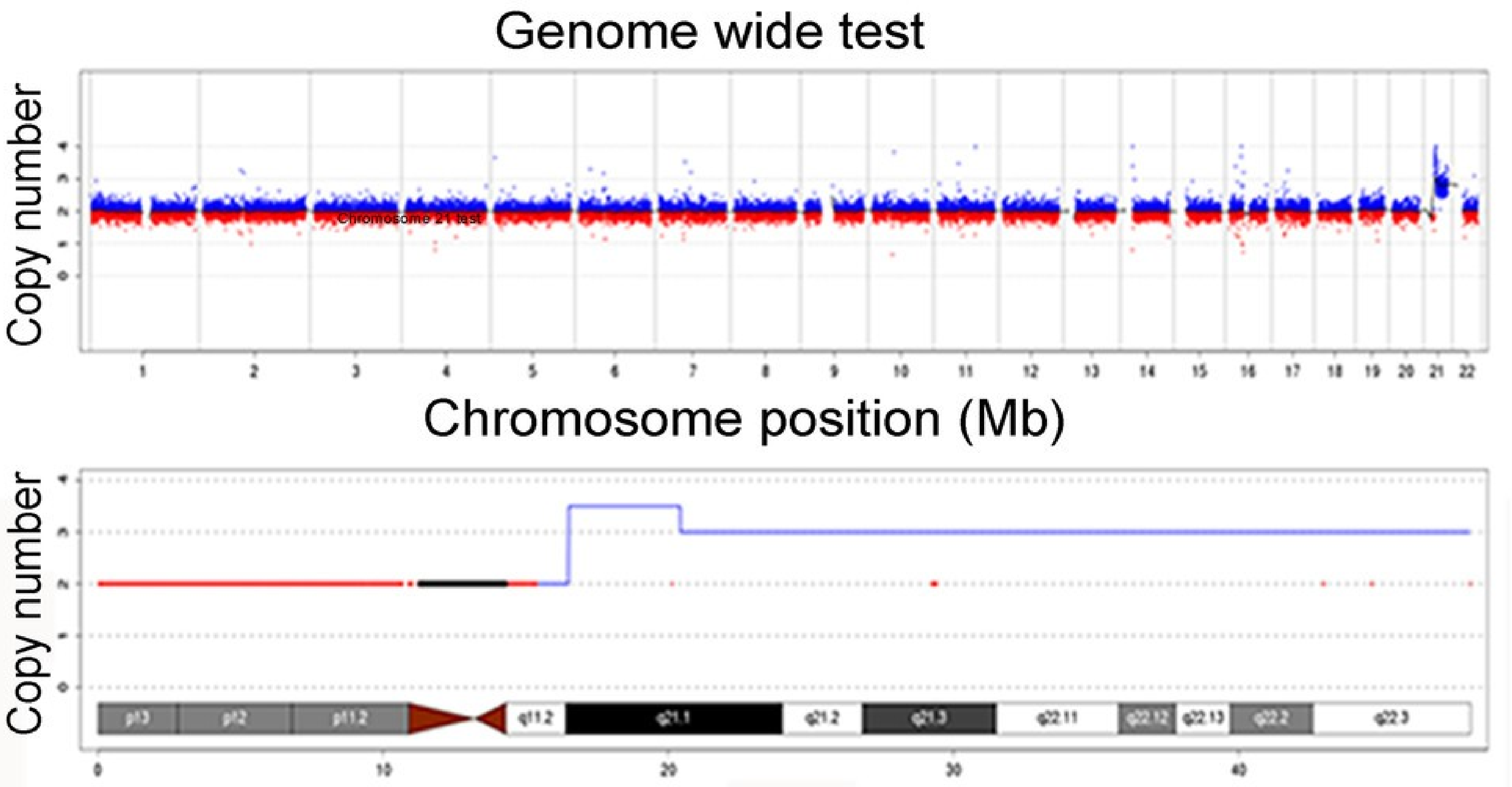
https://www.takarabio.com/learning-centers/next-generation-sequencing/technical-notes/reproductive-health-technologies/embgenix-pgt-a

Ví dụ 5: Sàng lọc bất thường nhiễm sắc thể ở thai nhi



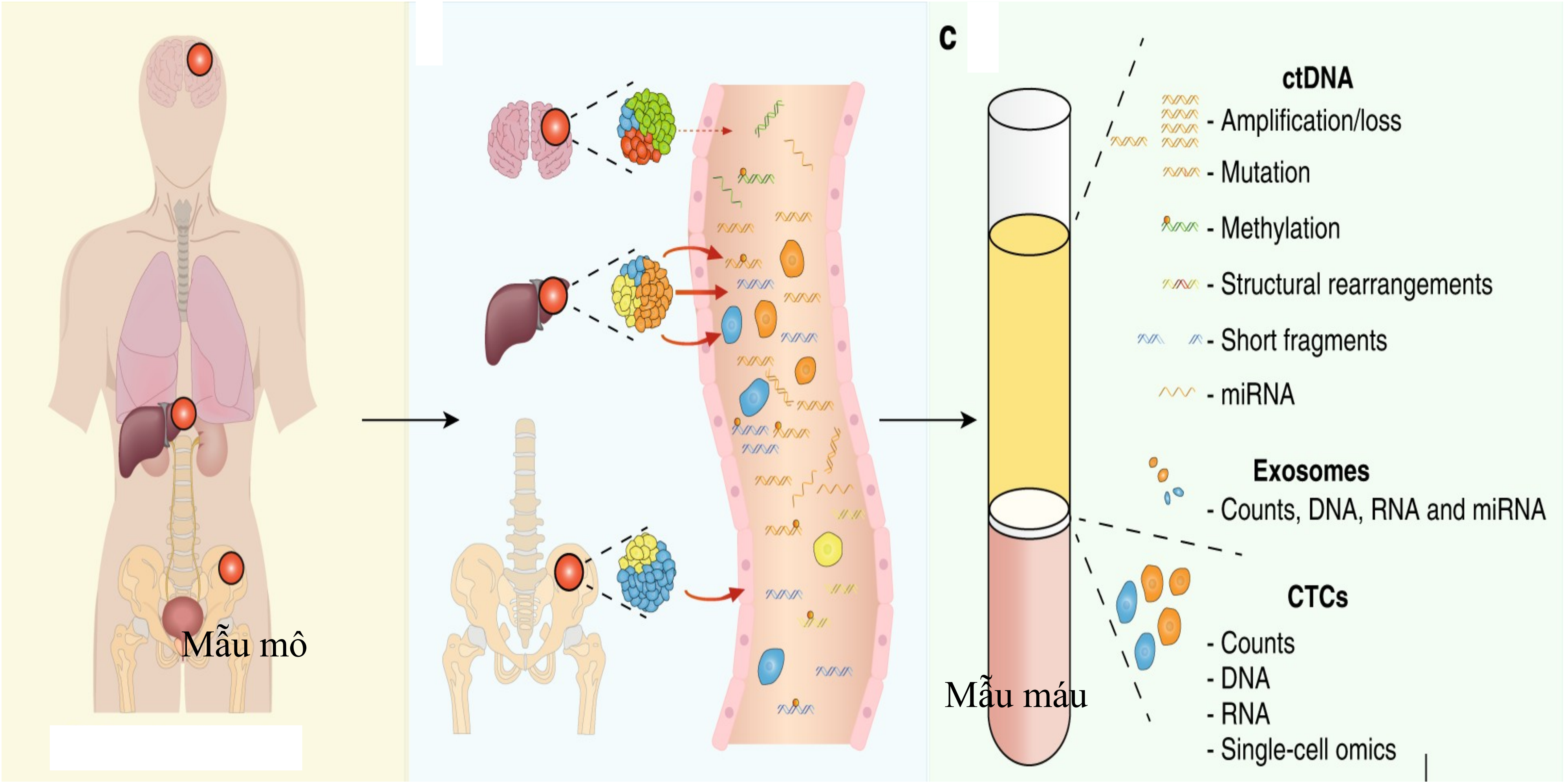
https://www.advancedwomensimaging.com.au/nipt/

Kết quả NIPT: Bất thường nhiễm sắc thể 21 ở thai nhi



10.7717/peerj.14400

Mẫu cho XÉT NGHIỆM gen: mẫu mô và máu

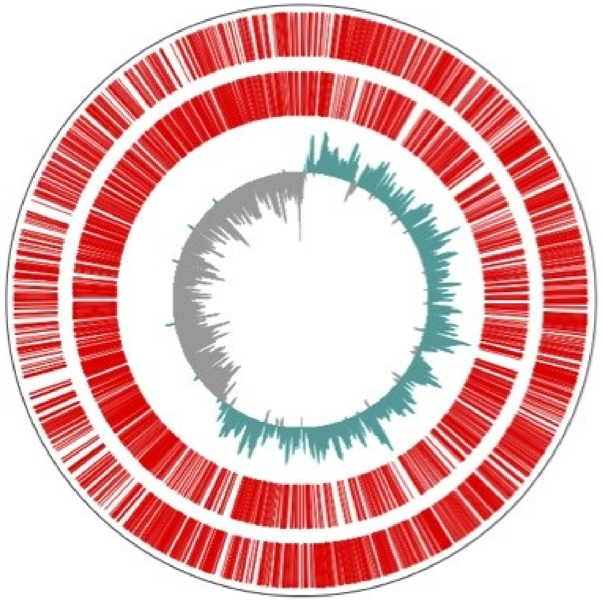


10.1038/s41416-022-01881-9

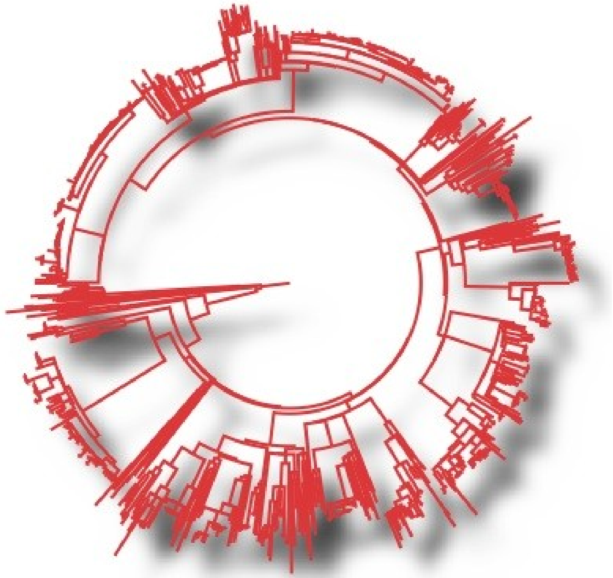
Ứng dụng NGS trong bệnh nhiễm

Phân tích bộ gen vi khuẩn: nghiên cứu một tác nhân gây bệnh ở mỗi mẫu phân lập

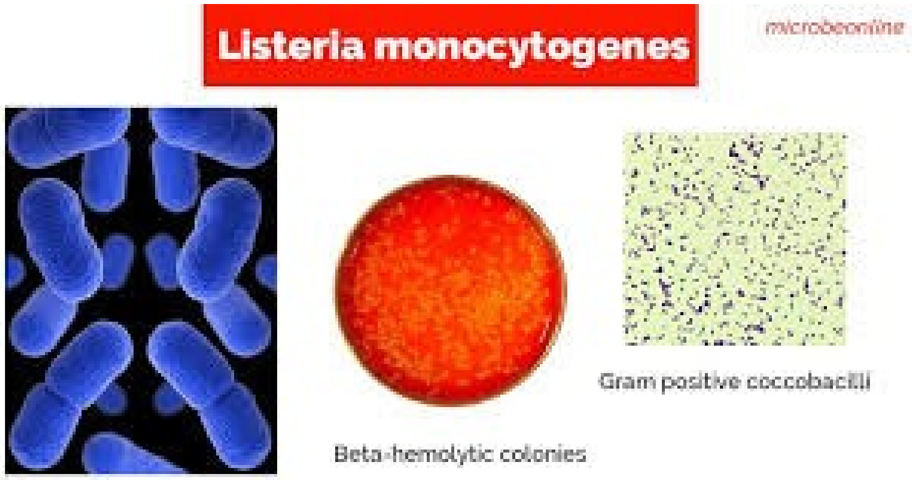
Phân lập

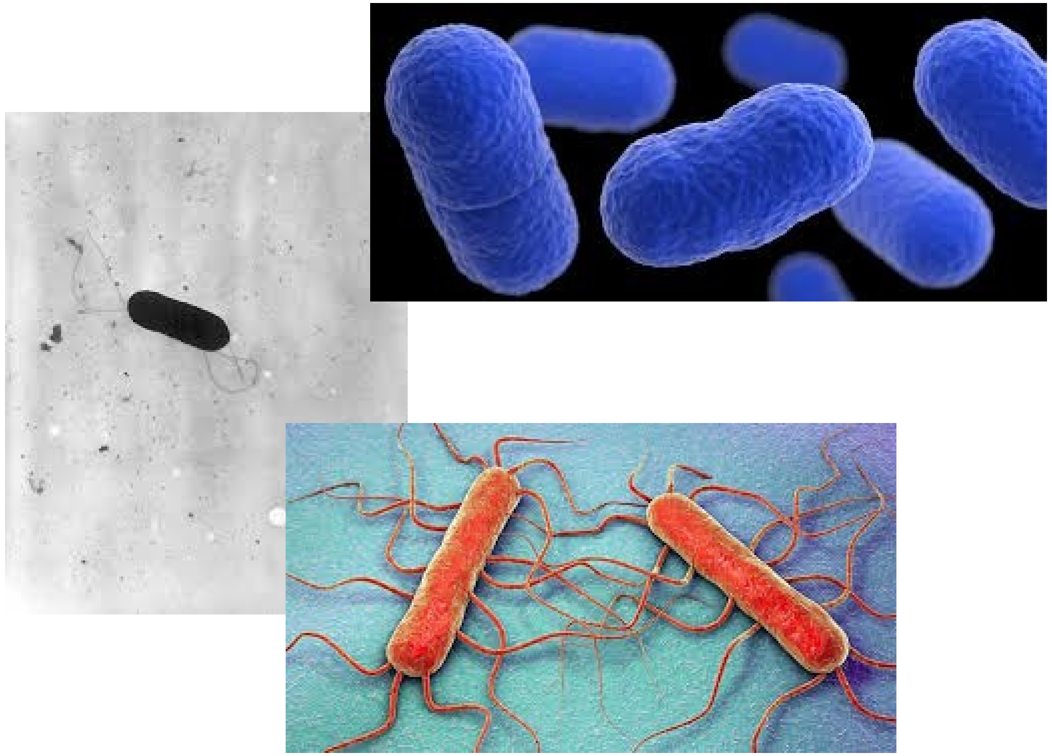


Giải trình tự bộ gen



Serotyping/genotyping





Listeria monocytogenes EGD-e, complete genome

GenBank: AL591824.1

GenBank   Graphics

```
>AL591824.1 Listeria monocytogenes EGD-e, complete genome
TAATTAAATCTAACAATTTCGTTACAGATTTCTTTACACACAAGTTATACACAAGTTAACTGGCTGTGGA
CAACCGTTTTTCACATCTGGACAGTTTTGTGGATAGATTTGGTAAGTCCTTGCTATCAGAGTGATTTTCT
GATATTATAATTCCTGTCGAATAGAAATATAGCTGGGGAAAACTAAAGTTATCCACAATACATTTTTACT
TTGTGGATAATTTTTTAACAGTGTTTGGATAACCTTATCCATAGCTTTTTCTATCTGTGGATAACTTTAT
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TTGGAAAACATATGCTTAATCCACGTTATGTTTTTGATACCTTTGTCATTGGTTCAGGGAACAGATTTGC
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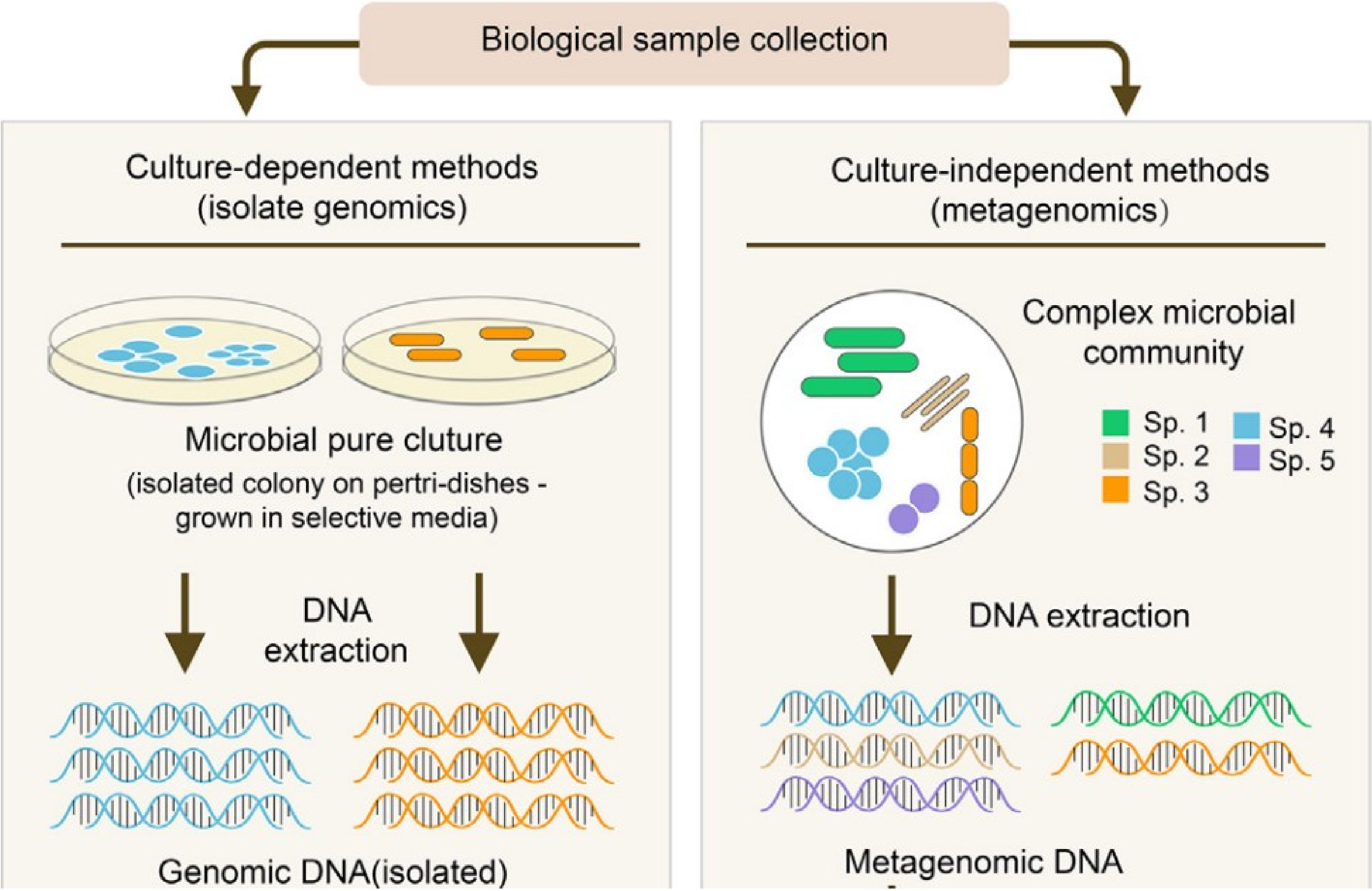
GTGTACAATGTGGATAACTGAAACATACTTACCCACAAGTTATCAACATGTGGAAAACTTTATGCAGCAT
GGCTTGTAACCTACTTATCCACAAATCCACAGCGCCTATTACTATTACTACGATTTTTTATTAATTAATT
```

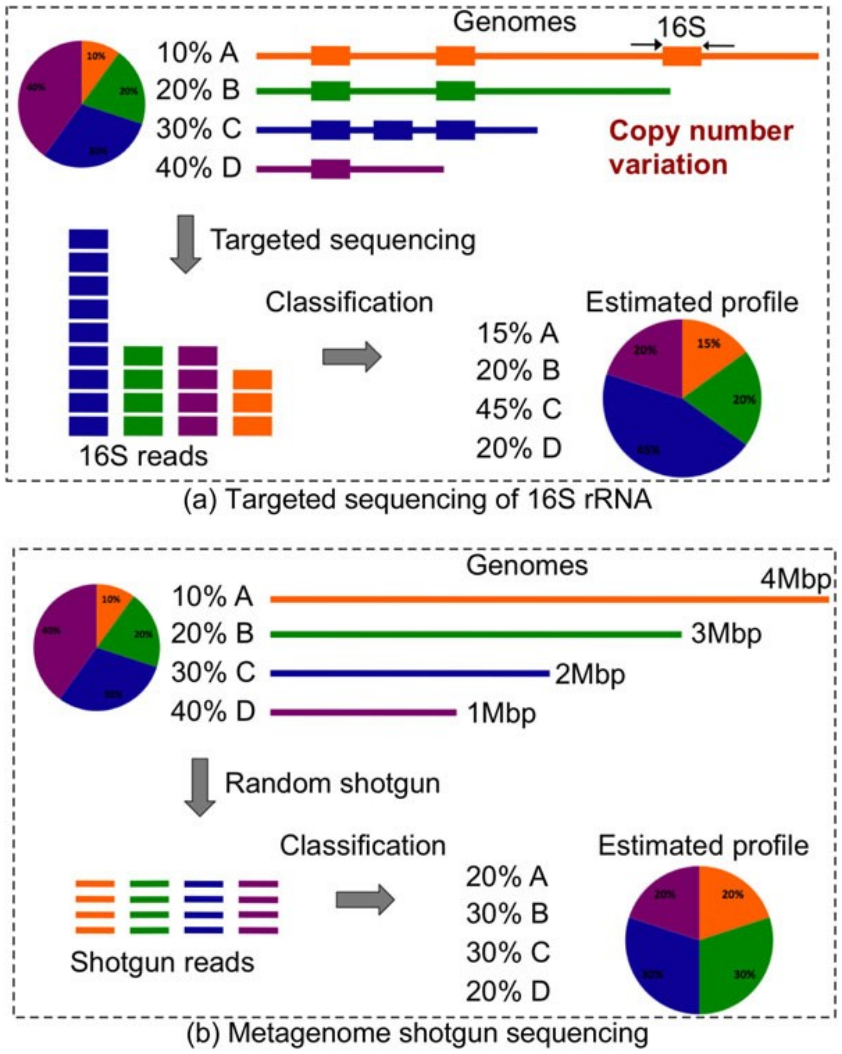
https://www.ncbi.nlm.nih.gov/nuccore/AL591824.1?report=fasta

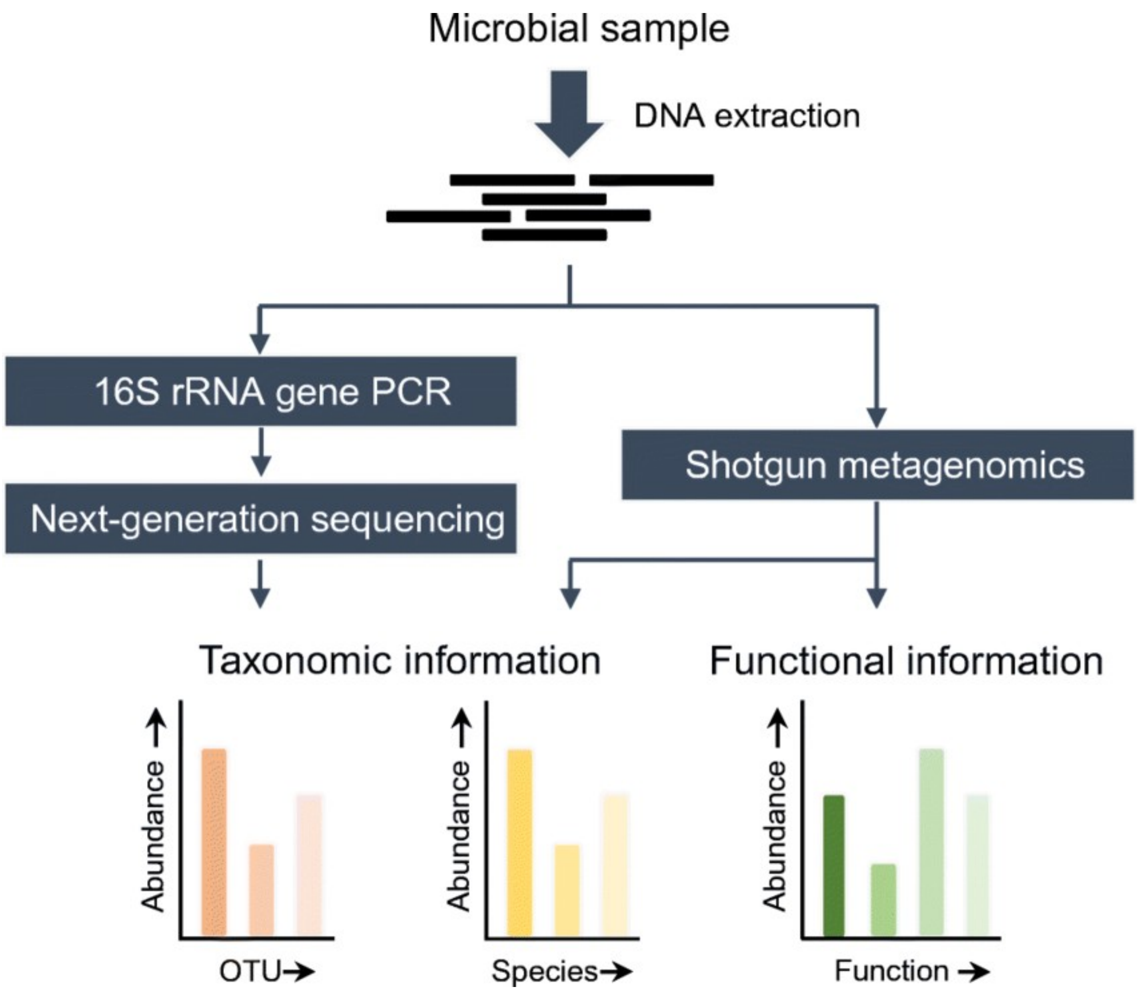
Phân tích một bộ gen vs nhiều bộ gen



10.1016/j.csbj.2021.11.028

Target vs Shotgun Microbiome Analysis





10.1186/1471-2164-12-S2-S4

10.1007/s10096-019-03520-3

Precision microbiome testing: **STI, HPV** and **AMR**

Women's Health Test

Patient Name:	nan	**Provider:**	Mony Sary, Yap Chew, Wendy Ullmer	**Patient ID:**	VS8
Gender:	nan	**Provider NPI:**	nan	**Specimen ID:**	KH20.45474
DOB:	nan	**Order Date:**	nan	**Specimen Type:**	nan
Age:	nan	**Health Status Reported:**	nan	**Collection Date:**	nan

Sexually Transitted Infections

Name	Associated Condition	Result
Neisseria gonorrhoeae	Gonorrhea, urethritis, pelvic inflammatory disease, gonococcemia, gonococcal ophthalmia neonatorum	Detected
Chlamydia trachomatis	Chlamydia, cervicitis, urethritis, pelvic inflammatory disease	Not Detected
Mycoplasma genitalium	Urethritis, cervicitis, pelvic inflammatory disease	Not Detected
Treponema pallidum	Syphilis	Not Detected
Haemophilus ducreyi	Chancroid	Not Detected
Trichomonas vaginalis	Trichomoniasis	Not Detected
Human papillomavirus	Cervical and anogenital cancers, genital warts	Detected
Herpes simplex virus	Genital herpes, oral herpes	Not Detected

Viruses Detected

Name	Associated Condition
Human papillomavirus 62 (HPV 62)	Unknown risk for cervical cancer

Note: Human papillomavirus (HPV) 16, 18, 31, 33, 35, 39, 45, 51, 56, 58, 59, and 68 are considered high-risk or probable high-risk due to their association with cervical cancer. HPV 6, 11, 42, 43, and 44 are considered low-risk for cervical cancer but may cause genital warts. Other HPV genotypes found in the sample may have intermediate or unknown risk for cervical cancer.

Antimicrobial Resistance Genes Detected

AMR Gene Name	Function	Drug Class
Neisseria.gonorrhoeae.folP	Dihydropteroate synthase (mutated)	Sulfonamide

Women's Health Test

Patient Name:	nan	**Provider:**	Yap Chew	**Patient ID:**	nan
Gender:	nan	**Provider NPI:**	nan	**Specimen ID:**	202122865
DOB:	nan	**Order Date:**	nan	**Specimen Type:**	nan
Age:	nan	**Health Status Reported:**	nan	**Collection Date:**	nan

Sexually Transitted Infections

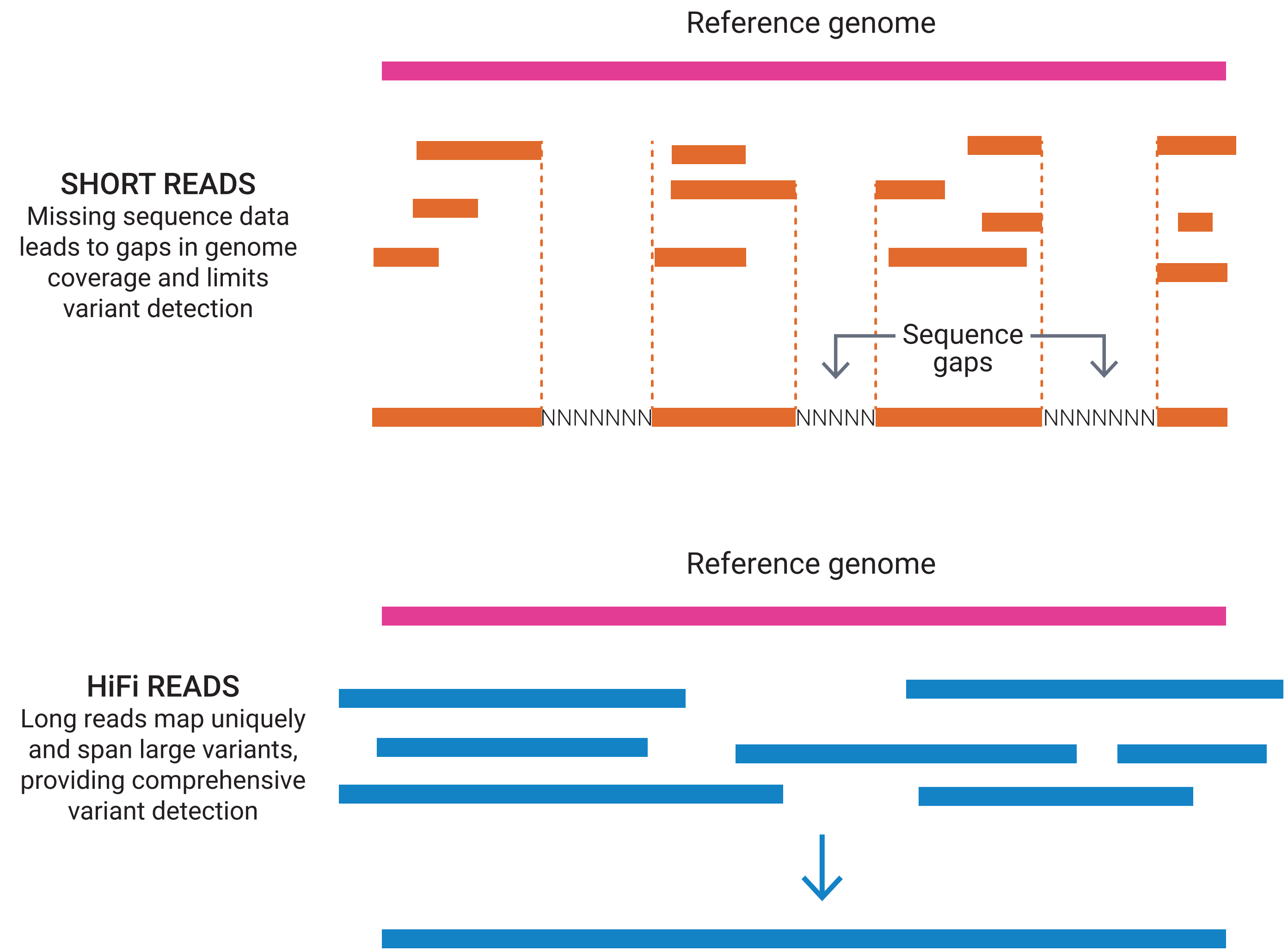
Name	Associated Condition	Result
Neisseria gonorrhoeae	Gonorrhea, urethritis, pelvic inflammatory disease, gonococcemia, gonococcal ophthalmia neonatorum	Not Detected
Chlamydia trachomatis	Chlamydia, cervicitis, urethritis, pelvic inflammatory disease	Not Detected
Mycoplasma genitalium	Urethritis, cervicitis, pelvic inflammatory disease	Not Detected
Treponema pallidum	Syphilis	Not Detected
Haemophilus ducreyi	Chancroid	Not Detected
Trichomonas vaginalis	Trichomoniasis	Not Detected
Human papillomavirus	Cervical and anogenital cancers, genital warts	Detected
Herpes simplex virus	Genital herpes, oral herpes	Not Detected

Viruses Detected

Name	Associated Condition
Human papillomavirus 52 (HPV 52)	High-risk for cervical cancer
Human papillomavirus 68 (HPV 68)	High-risk for cervical cancer

Note: Human papillomavirus (HPV) 16, 18, 31, 33, 35, 39, 45, 51, 56, 58, 59, and 68 are considered high-risk or probable high-risk due to their association with cervical cancer. HPV 6, 11, 42, 43, and 44 are considered low-risk for cervical cancer but may cause genital warts. Other HPV genotypes found in the sample may have intermediate or unknown risk for cervical cancer.

Third Generation Sequencing – PacBio Long Read



https://www.youtube.com/watch?v=7yYPHatccgw

Third Generation Sequencing – PacBio Long Read

Feature	PacBio Long-Read Sequencing	Short-Read Sequencing
Read Length	1 kb to >20 kb	50 to 300 bp
Primary Strength	Mapping complex regions, structural variants, and long repeats.	High depth, precision, and low-cost SNP detection.
Genome Assembly	Exceptional assembly continuity with fewer gaps.	Requires complex computational assembly, struggles across repeats.
Epigenetics	Can detect native base modifications (like DNA methylation) in real-time.	Loses methylation data during amplification.
Cost	Higher cost per base.	Lower cost per base.

https://www.youtube.com/watch?v=7yYPHatccgw

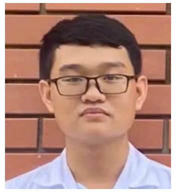
The Bioinformatics and Data Science Team at Precision Gene

Hong Ong, Msc.

Nguyen Le Duc Minh, Dr.
Bioinformatic Specialist

Ngo Dai Phu, Msc.
Project Manager

Dao Khuong Duy, Msc.
Bioinformatic Specialist



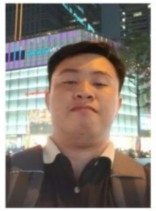
Phan Phát
Bioinformatic Specialist

Tân
Bioinformatic Specialist

Giau Le, Bsc.
Software Engineer

Ho Phu Quy, Bsc.
Bioinformatic Specialist

Pham Hieu Dan, Bsc.
Bioinformatic Specialist



Nguyen Manh Hung, Bsc.
Bioinformatic Specialist

Thanh Nguyen, Bsc.
Bioinformatic Specialist

Nguyen Quang Khai, Bsc.
Bioinformatic Specialist

Kim Van, Bsc.
Bioinformatic Specialist

Huy Ha, Bsc.
Bioinformatic Specialist

Nguyen Thanh Cong, Msc.
Bioinformatic Trainee

Xin chân thành cảm ơn!

Luu Phuc Loi, PhD

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